FILTER BUBBLES and ECHO CHAMBERS

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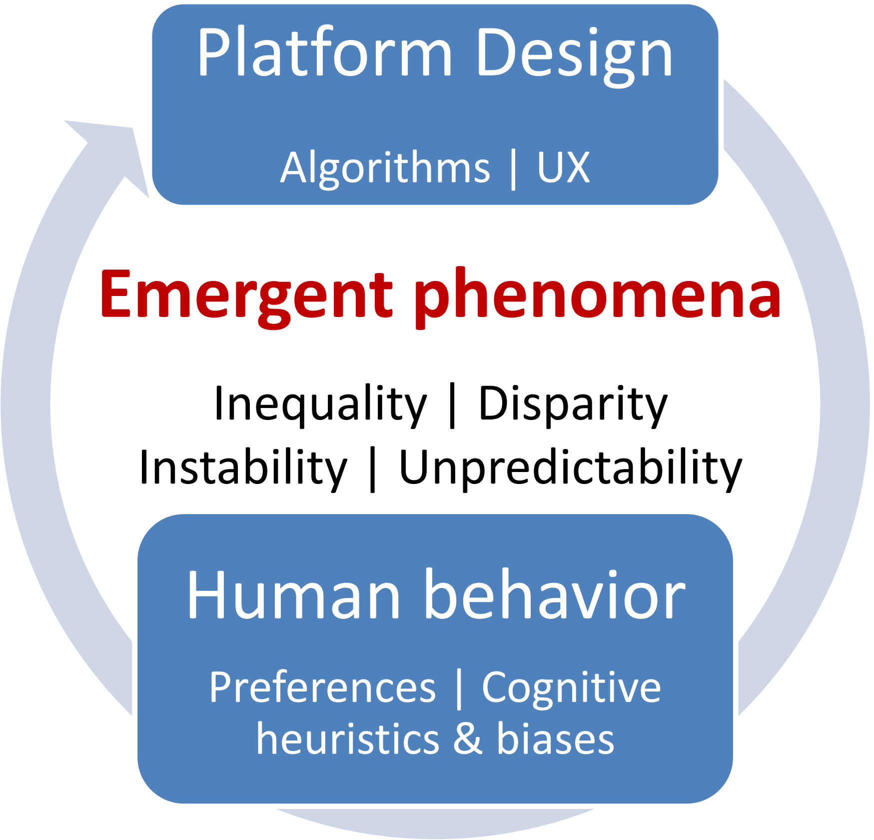
DSCI 531

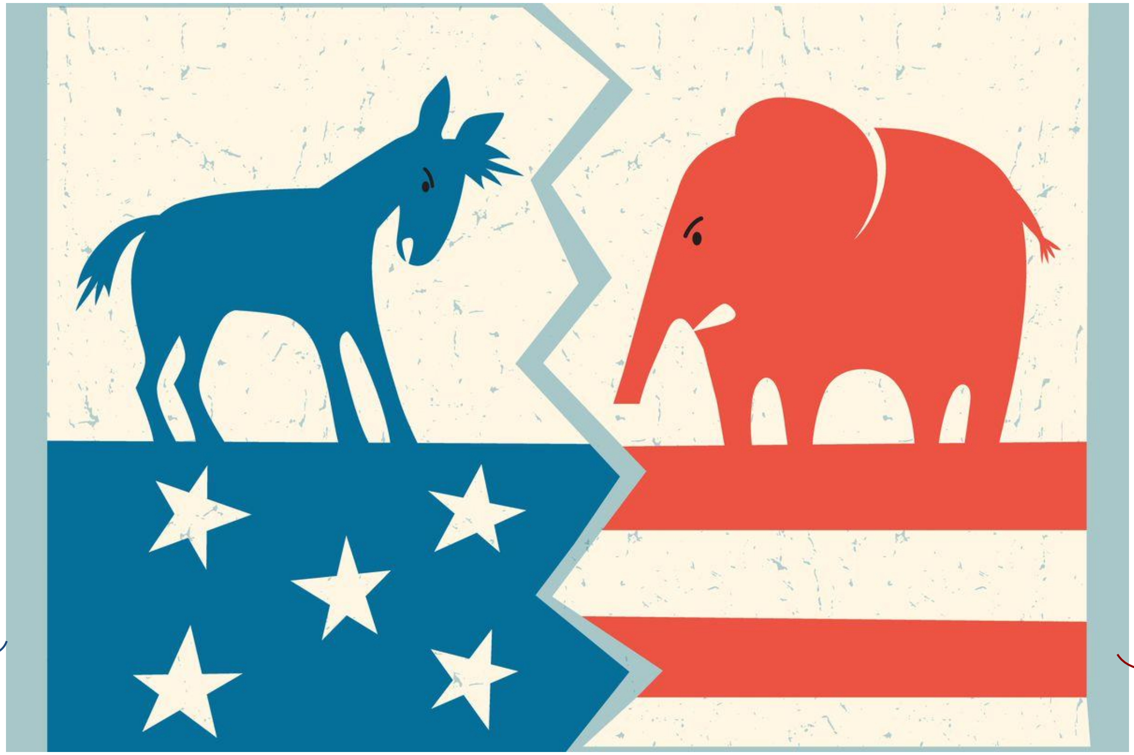
Dynamics of human-algorithm systems

- The intricate **interplay** between **algorithms** and **people** within online platforms is reshaping social interactions, leading to **emergent, unpredictable** behaviors
- What happens to social systems?



Polarization

"...When MAGA republicans are **destroying Democracy**..." - Joe Biden



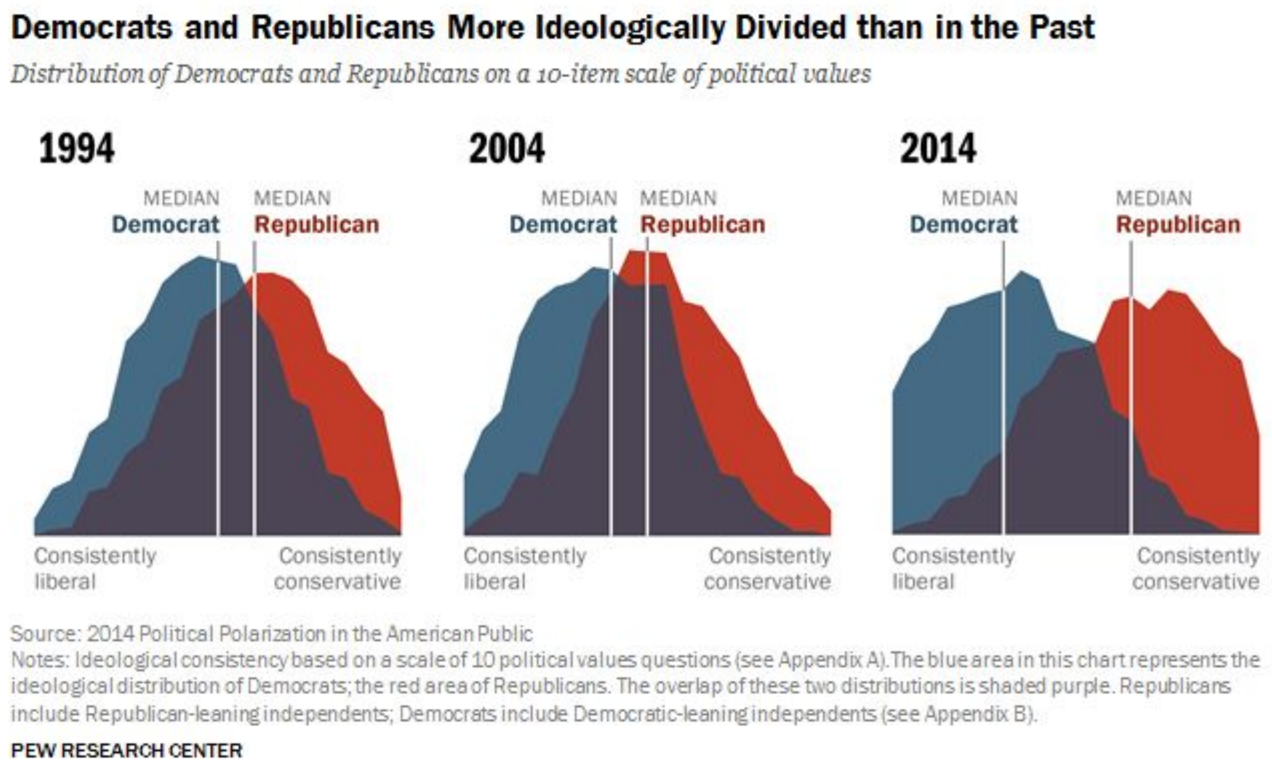
"...if these far-left politicians ever get into power they will **demolish** not only your industry but the **entire US economy**.." - Donald Trump

Image taken from UPenn Today

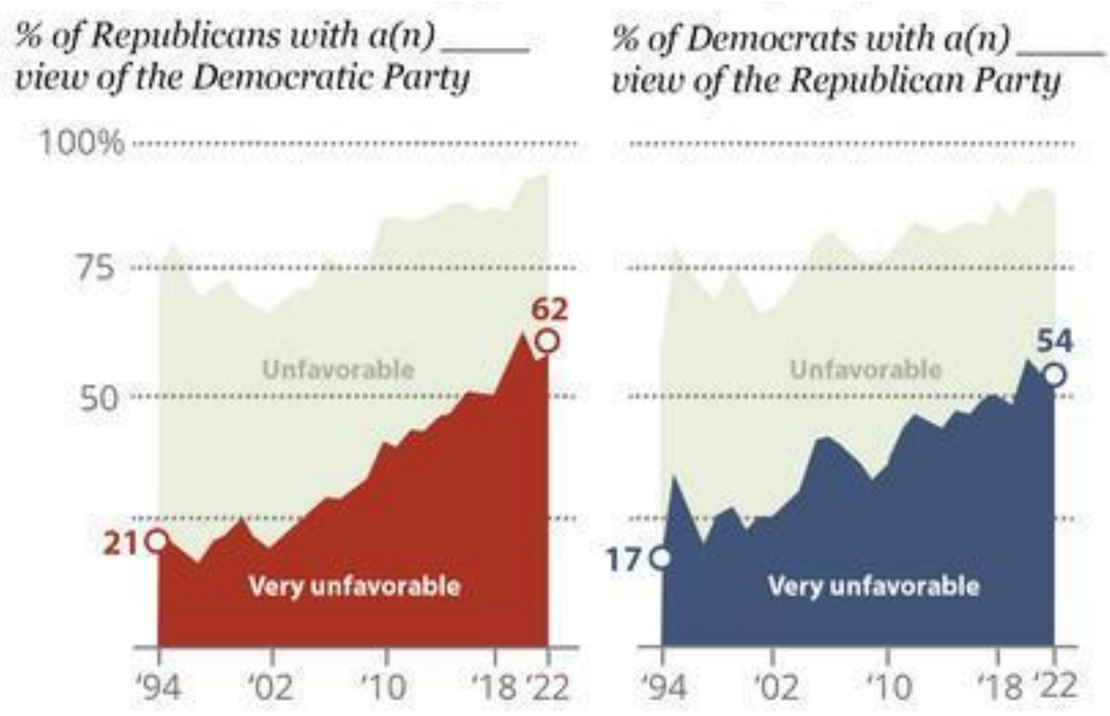
***Polarization*: divergence of attitudes dividing a population** into two groups with **sharply contrasting opinions** or beliefs

*Bavel et al. **How social media shapes polarization**. Trends in Cognitive Sciences (2021)

Democrats and Republicans disagree more than ever



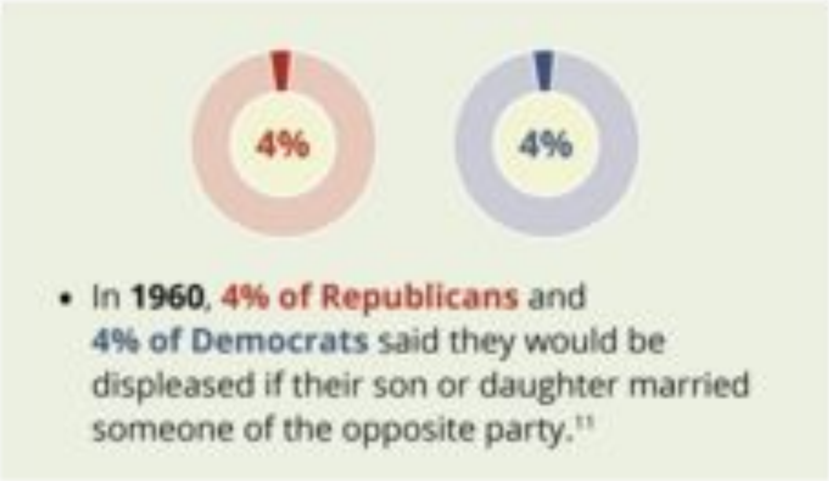
Rising partisan distrust

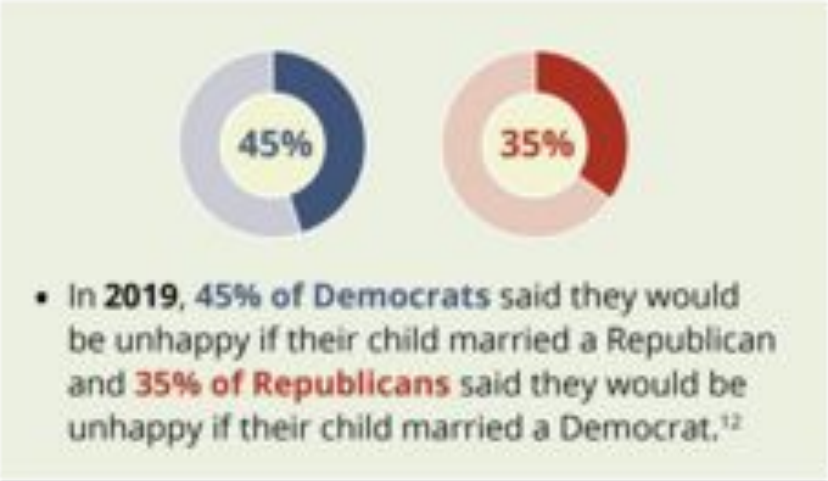


Note: Based on partisans and does not include those who lean to each party.
Source: Yearly averages of survey data from Pew Research Center American trends Panel (2020–2022) and Pew Research Center phone surveys (1994–2019).

PEW RESEARCH CENTER

… influencing lifestyle choices



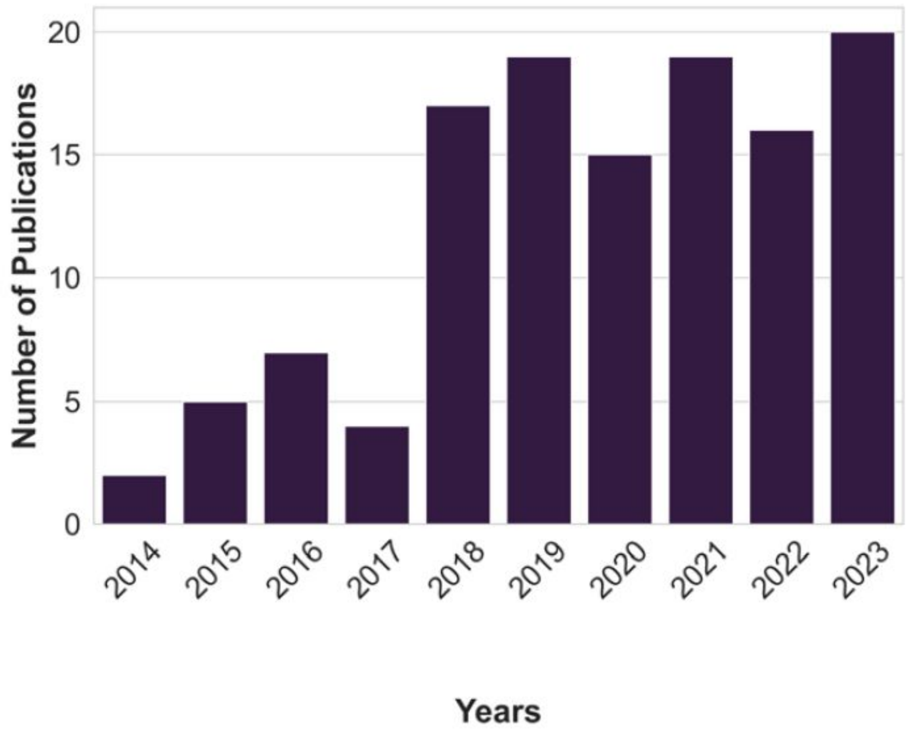


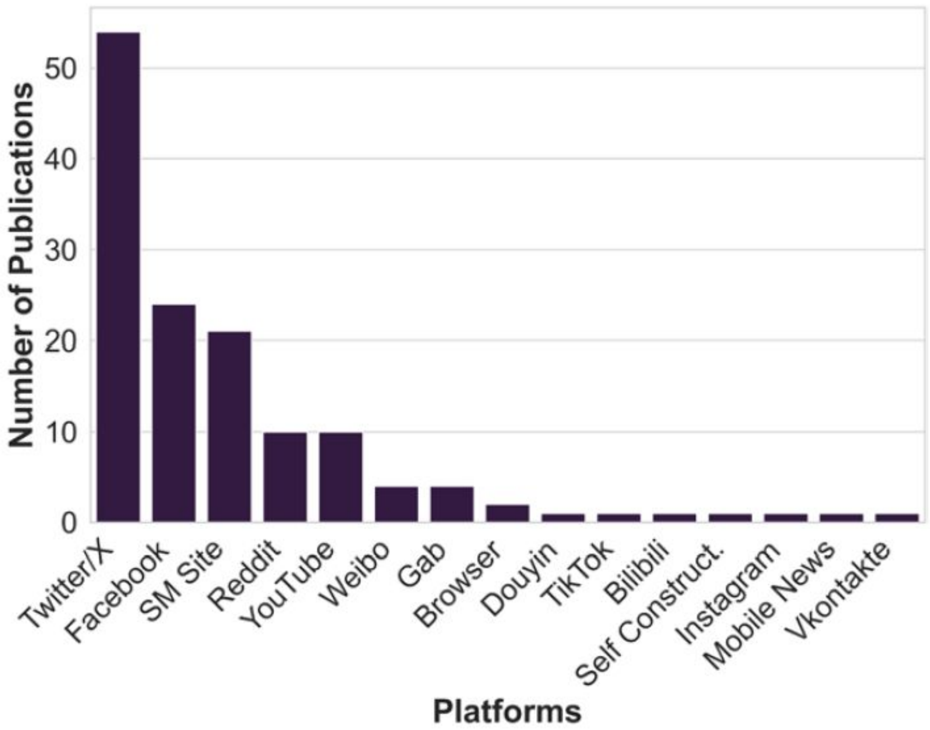
CREDIT: Facing History & Ourselves (unrestricted)

Social media as a tool of societal discord

- **Social media** platforms shapes social interactions by curating and recommending content. Emergent phenomena include
  - **Filter bubbles**: Personalization isolates users from diverse viewpoints.
  - **Echo chambers**: Users mostly interact within like-minded communities.
  - **Emotional manipulation**: Platforms amplify emotional, especially *outrage-driven*, content to maximize engagement.
- These mechanisms together intensify polarization and distort psychological states.
  - **Affective polarization**: Ideological groups distrust and dislike each other
  - **Peer contagion**: psychological wellbeing is compromised

Popular research topic



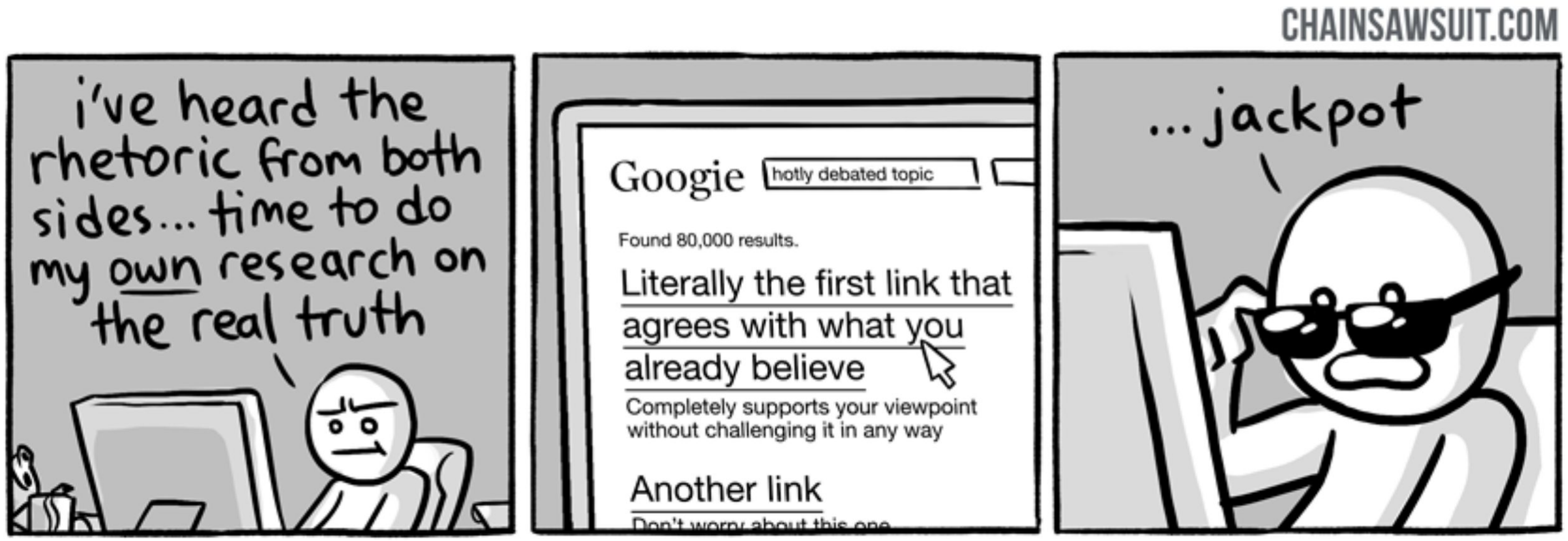


Hartmann *et al.* (2025) A systematic review of echo chamber research: comparative analysis of conceptualizations, operationalizations, and varying outcomes. *J Comput Soc Sc*.

Filter bubbles

A filter bubble is a state of intellectual isolation caused by **content personalization algorithms** [Eli Pariser (2011). "The Filter Bubble"]

- Personalization algorithms create invisible filters by prioritizing content aligned with user preferences and past behavior
    - Search algorithms, Social media feed algorithms
  - Users are shown what they want to see
- Potential harms
  - Confirmation bias
  - Users are isolated from alternative viewpoints, reinforcing existing beliefs
  - Undermines discourse by fragmenting shared public knowledge


CHAINSAWSUIT.COM
i've heard the rhetoric from both sides... time to do my own research on the real truth
Googie
hotly debated topic
Found 80,000 results.
Literally the first link that agrees with what you already believe
Completely supports your viewpoint without challenging it in any way
Another link
...jackpot

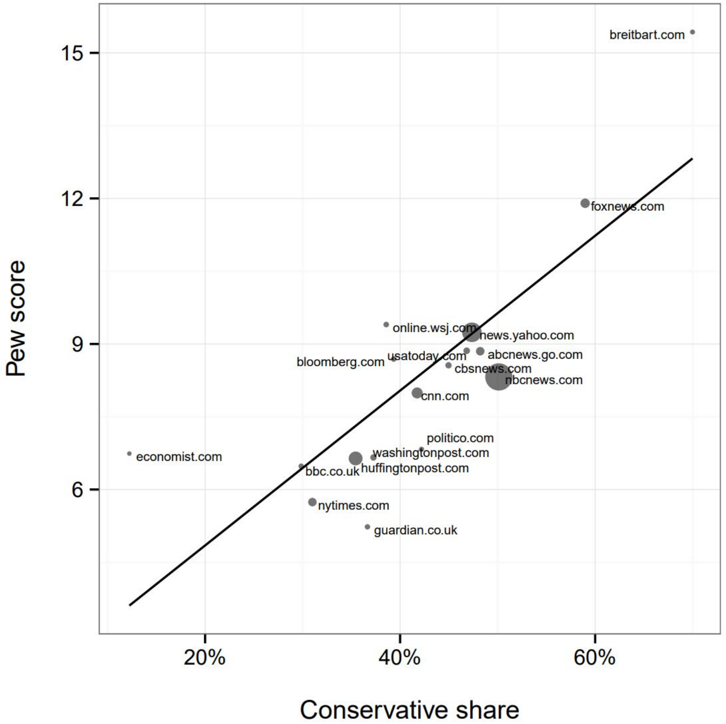
Open questions

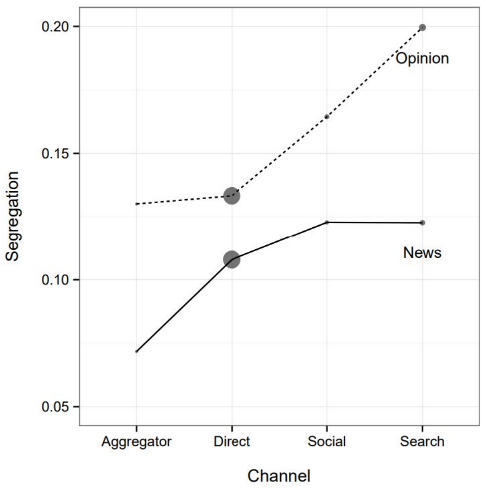
- Do filter bubbles exist?
- Are they harmful?
- Do people want them?
- … Evidence is mixed!

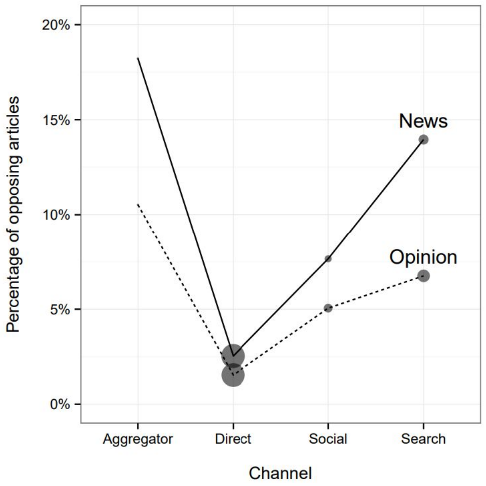
Filter bubbles in news media

Does online news consumption affect political polarization?

- Browser data from 50,000 users; compared direct visits, search, and social media referrals [Flaxman, Goel & Rao (2016) *Public Opinion Quarterly*]
  - Social media and search increase ideological segregation relative to direct visits
  - Users exposed to more one-sided views via social media
  - However, they also consume more total news

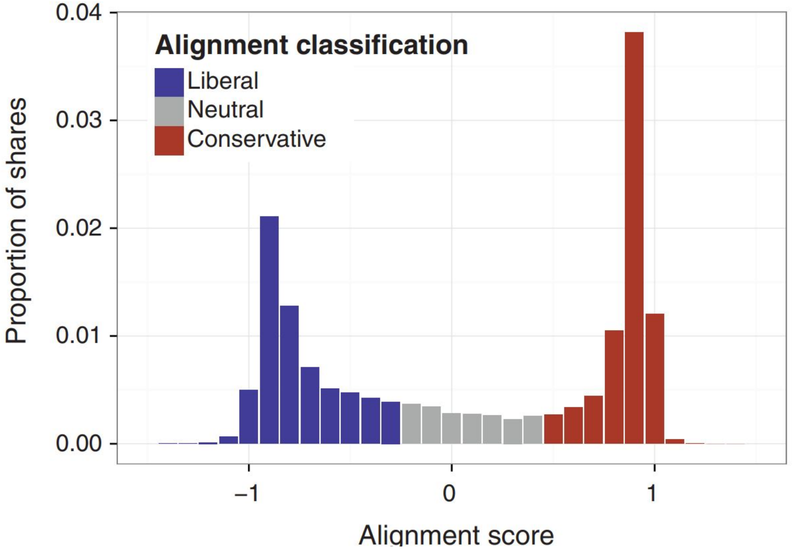






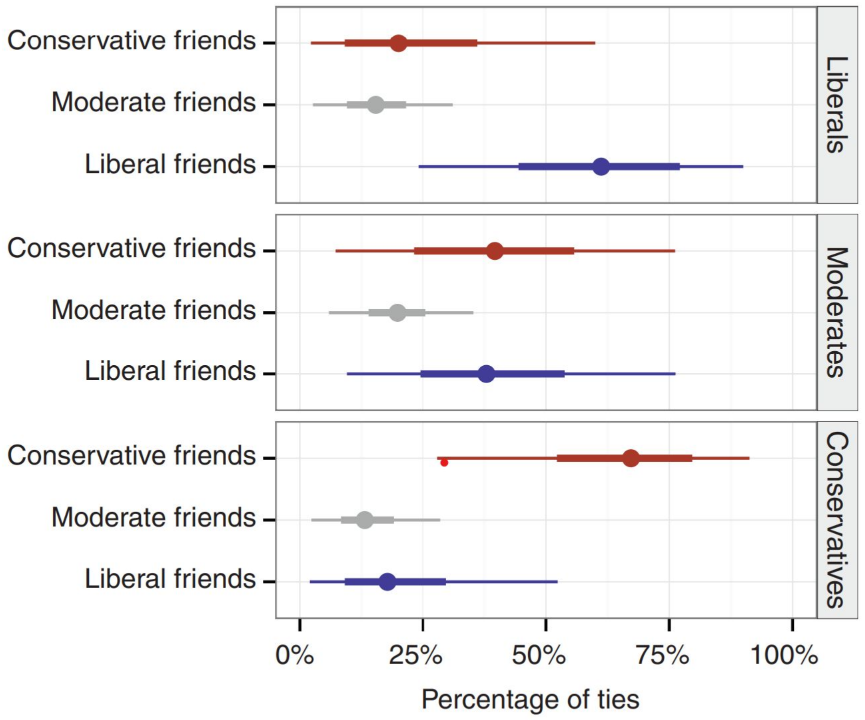
Is Facebook a filter bubble?

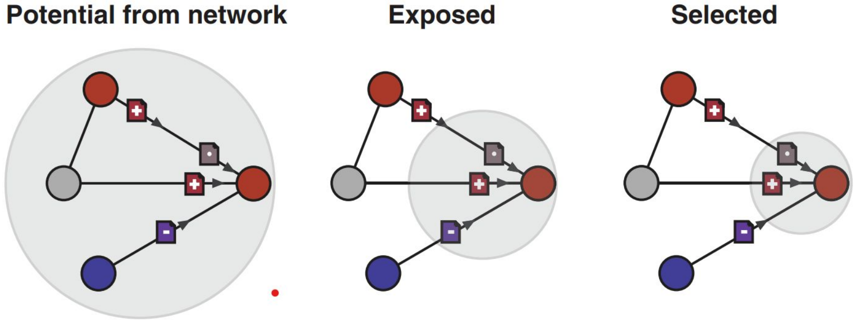
- Do Facebook algorithms reduce exposure to ideologically diverse content?
- Field experiment reducing users' Facebook time reduce political polarization
- Internal data from Facebook contradicts this view [Bakshy, Messing & Adamic (2015), *Science*]
  - Analyzed user behavior and algorithmic exposure on Facebook
    - Facebook users encounter some cross-cutting content
    - However, individual choices play a larger role than the algorithm in limiting diversity
    - Conclusion: Personal behavior contributes more to filter bubbles than Facebook's algorithm

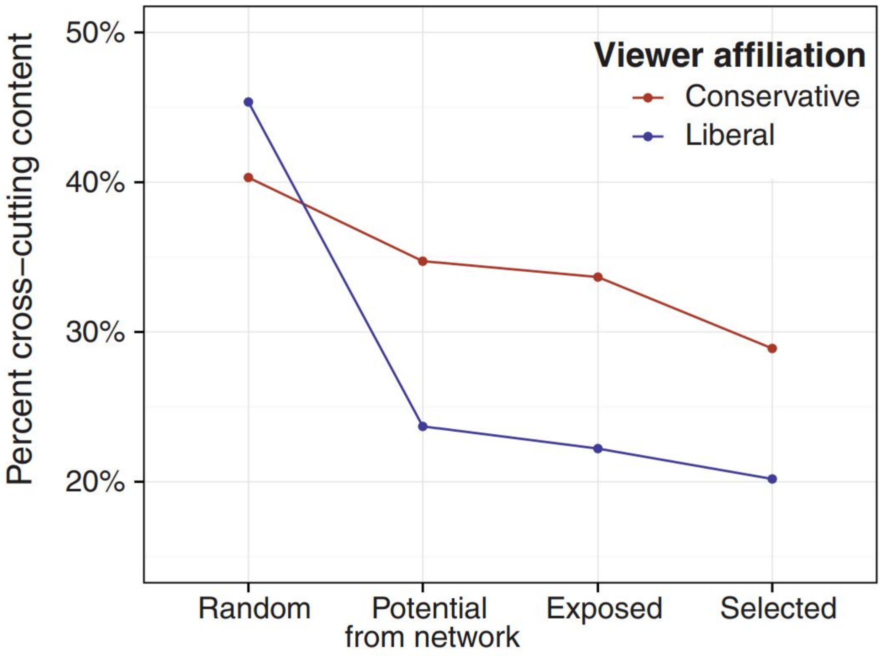


Users exposed to crosscutting content but select ideologically aligned content

Homophily in self-reported ideological affiliation. Proportion of links to friends of different ideological affiliations for liberal, moderate, and conservative users.







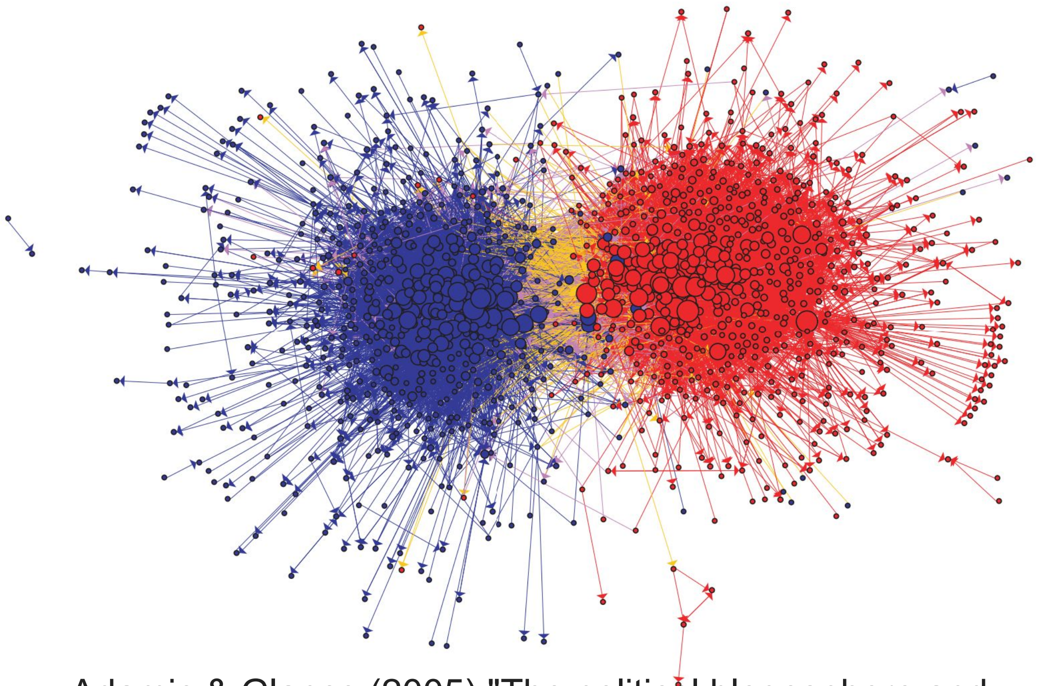
Are filter bubbles harmful?

- Users prefer ideologically aligned content when given control over news customization
  - Self-selected personalization
- But, while filter bubbles exist, their impact is often overstated [Zuiderveen Borgesius, (2016). *Internet policy review*]
- People are still exposed to diverse views through multiple channels

Echo chambers

- Echo chambers are environments where users are exposed predominantly to information that reinforces their existing views.
- Social media platforms amplify this by connecting like-minded individuals and filtering content.

Community structure of political blogs of 2004 election. The colors reflect political affiliation: conservative (red), and liberal (blue). Orange links go from liberal to conservative, and purple ones from conservative to liberal.



Adamic & Glance (2005) "The political blogosphere and the 2004 US election: divided they blog."

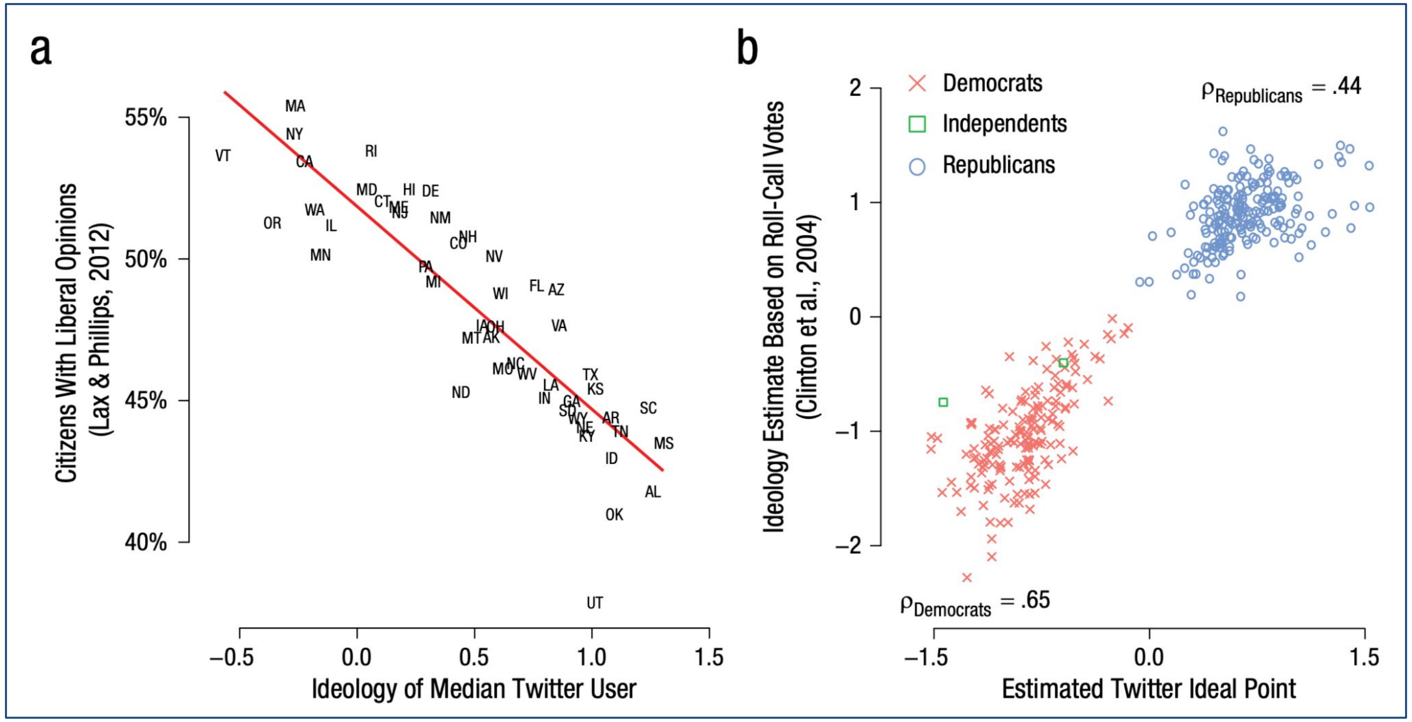
Historical view

- Echo chambers are information cocoons (Cass Sunstein (2001, 2017). *Echo Chambers*, *#Republic*)
- Echo chambers facilitate group interactions that reinforce extreme views
  - People seek information that confirms their beliefs (information cocoons).
  - Group discussions among like-minded individuals shift opinions toward extremes (group polarization).
  - Democracy requires exposure to diverse viewpoints, which echo chambers undermine.

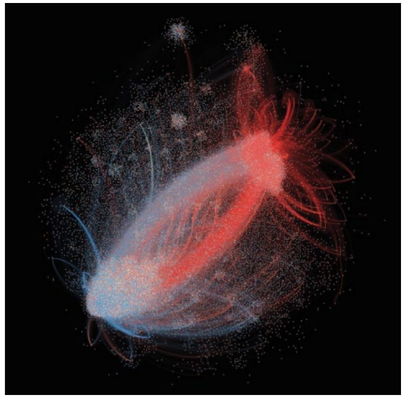
Is Twitter an echo chamber?

Network Based Approach: Friend/Follow Relationships

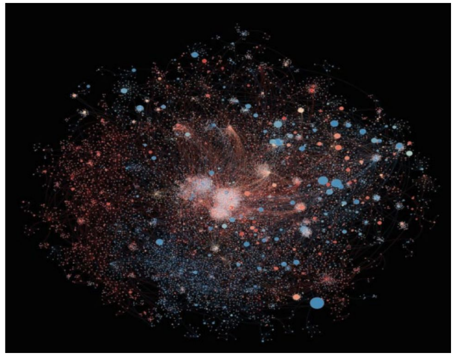
- Users are somewhat exposed to diverse viewpoints.
- However, communication patterns still reflect ideological clustering.
- Echo chambers are real and prevalent, especially in political discussions, amplifying ideological segregation.



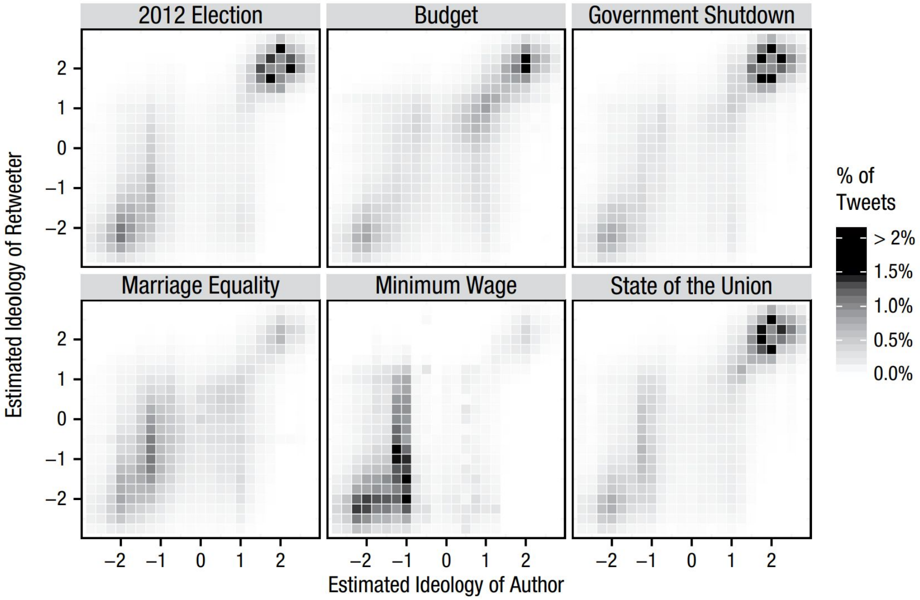
Barbera et al. Tweeting from Left to Right: Is online political communication more than an echo chamber? *Psychological Science* (2015)

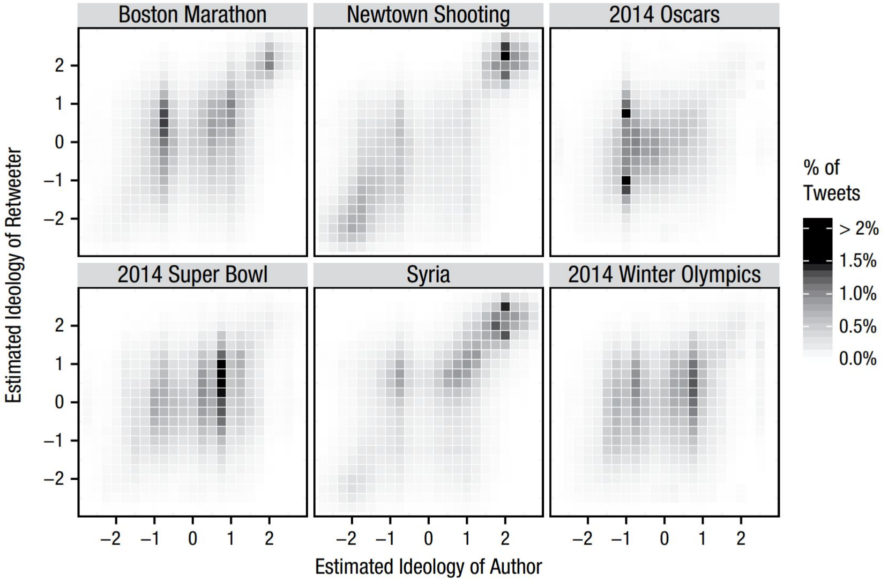


Retweet network of 2012 elections



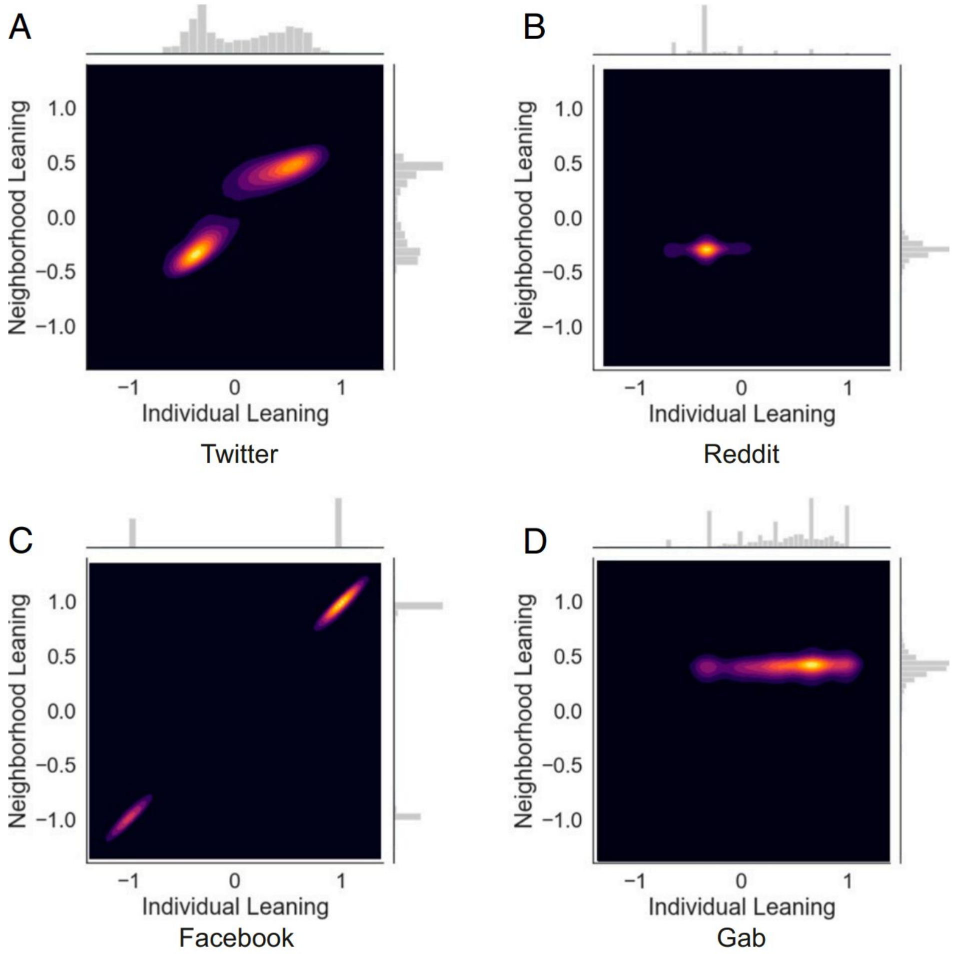
Retweet network of 2014 Winter Olympics





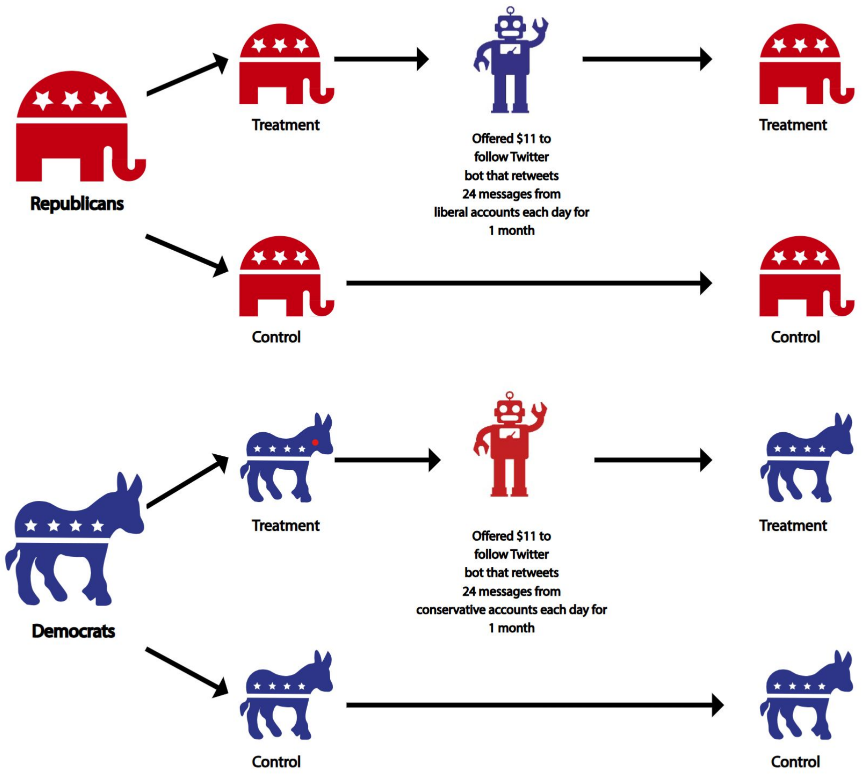
How strong are echo chambers on other platforms?

- Comparative study of Facebook, Twitter, Reddit, Gab (Cinelli et al 2021)
  - Facebook shows the strongest ideological segregation.
  - Reddit and Twitter show weaker echo chamber effects.
  - Echo chambers exist, especially in political discussions, but differ by platform architecture.



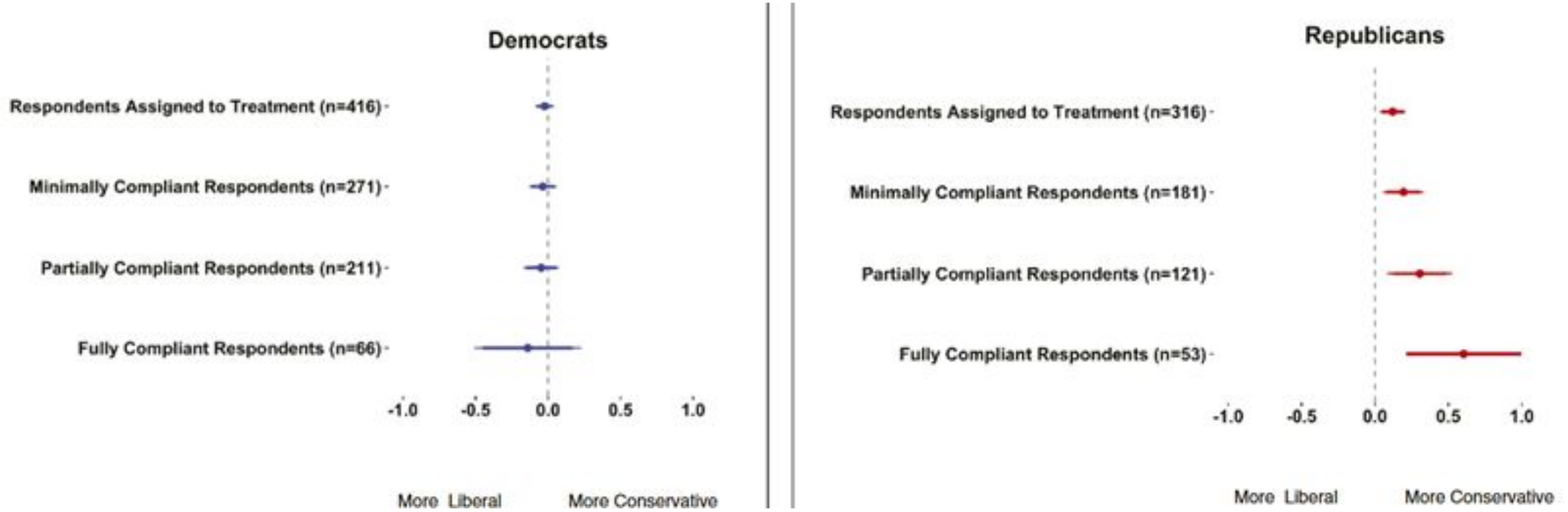
Does breaking echo chambers reduce polarization?

- Randomized field experiment on Twitter [Bail (2018)]
  - Participants were offered financial incentives to follow a Twitter bot that retweeted messages from elected officials and opinion leaders from the opposite political party
  - Experiment ran for one month during 2017
  - measured political attitudes before and after the intervention
  - used a control group for comparison



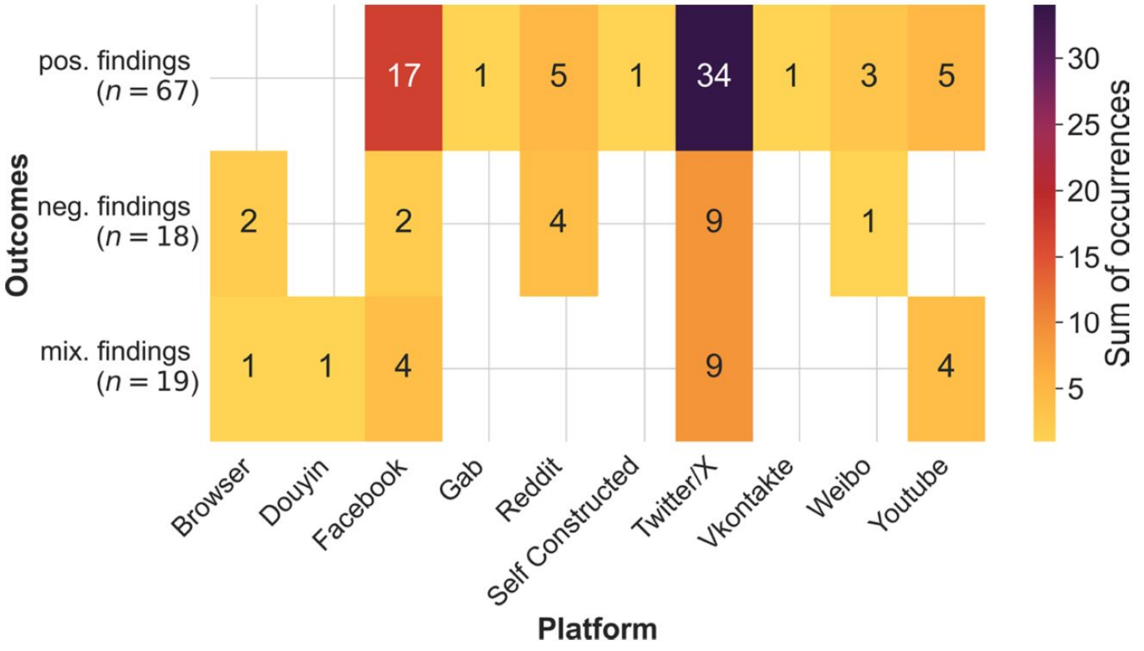
Backfire effect

- Exposure to opposing views **increased** polarization
- Republicans who followed a liberal Twitter bot became significantly more conservative in their views
- Democrats who followed a conservative Twitter bot showed slight increases in liberal attitudes, though these were not statistically significant
- Exposure to opposing views did not increase understanding of the other side's perspectives



Open questions

- Are echo chambers an inevitable outcome of online communication?
- How can platforms incentivize cross-cutting exposure without backfire?
- Should interventions focus more on users or algorithms?
- Is it possible to "burst" echo chambers effectively?



Heatmap of coded outcomes in echo chamber studies per researched platform. The colors indicate the total number of occurrences.

Hartmann *et al.* (2025) A systematic review of echo chamber research. *J Comput Soc Sc*.

Emotional manipulation in social media

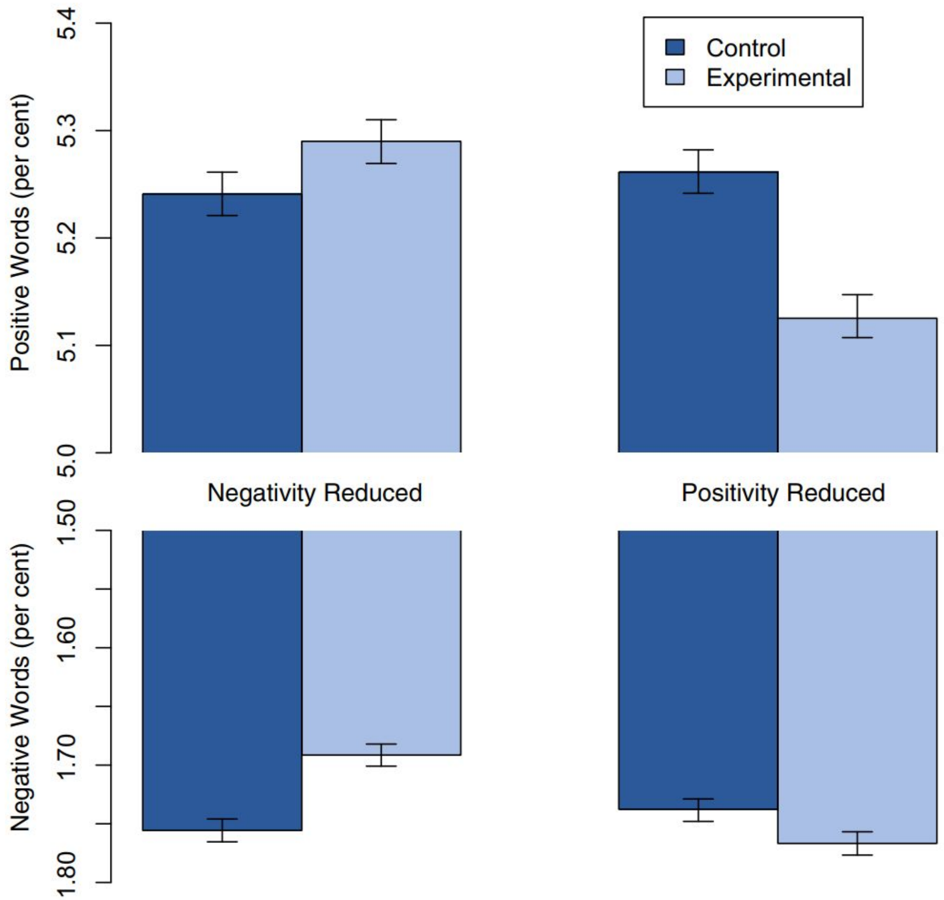
Platforms exploit emotional responses to shape user behavior

- **Engagement Optimization**: Algorithms maximize time on platform by favoring emotional content.
- **Moral Outrage**: Moralized anger is highly viral and encouraged by platform design.
- **Partisan Reinforcement**: Partisan emotional content is more likely to be reshared, entrenching division.
- **Emotional Contagion**: Emotional tone of feeds can influence users' own emotions and behavior.

Facebook Emotion Contagion Experiment

Can emotions spread through social networks without direct interaction?

- Method [**Kramer, Guillory & Hancock, 2014, PNAS**]
  - Experimentally reduced positive or negative emotional content in users' news feeds.
  - Users exposed to less positive content posted more negative updates.
  - Demonstrated "emotional contagion" through social networks.
  - Raised major ethical concerns about manipulation without consent.

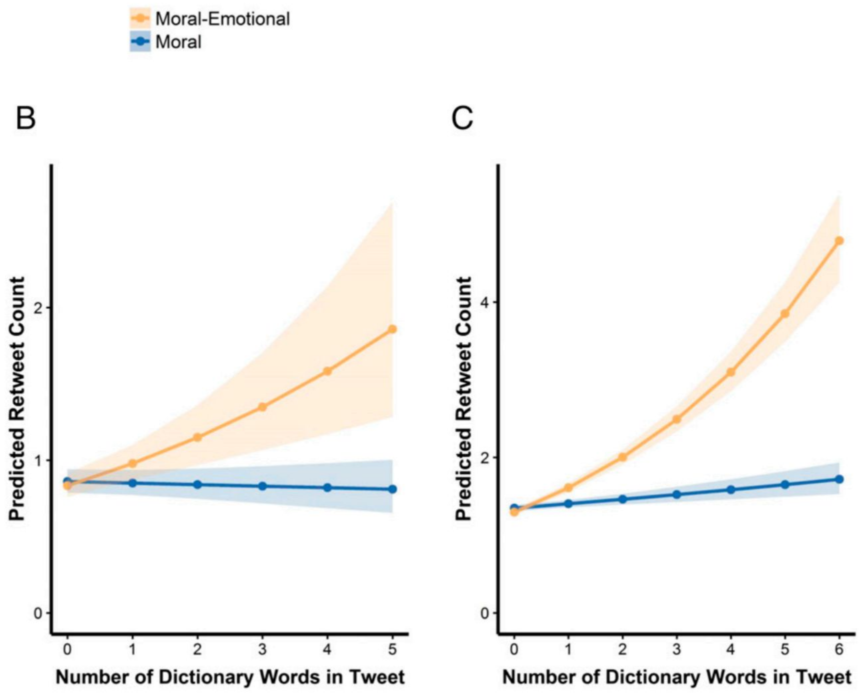


Mean number of positive (Upper) and negative (Lower) emotion words (percent) generated people, by condition. Bars represent standard errors.

Twitter Emotion Contagion

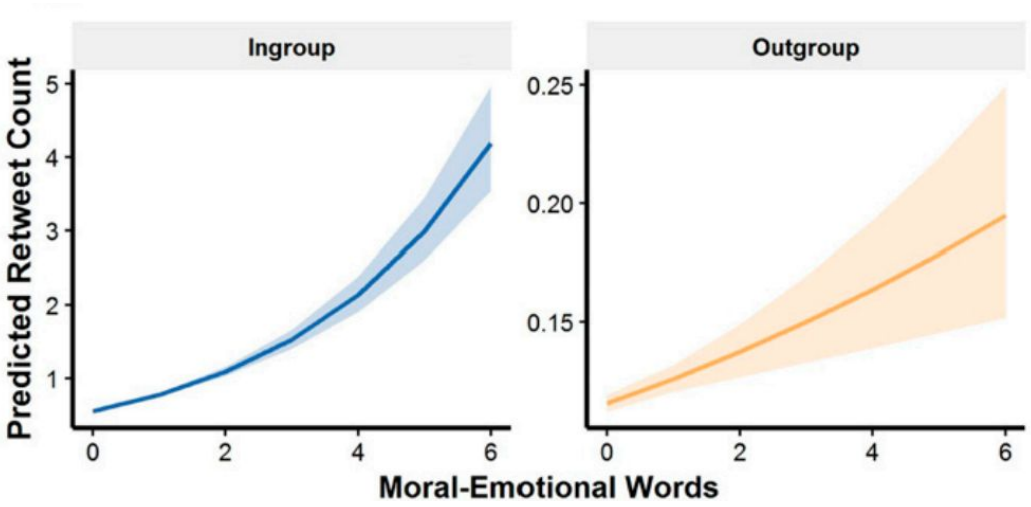
Does emotional language increase content sharing on social media?

- **Method**:
  - Analysis of millions of tweets about controversial topics.
  - Posts containing moral and emotional words (e.g., "hate," "evil") were shared more widely.
  - Emotional arousal boosts the diffusion of political content.
  - Moral outrage is a particularly viral form of communication.

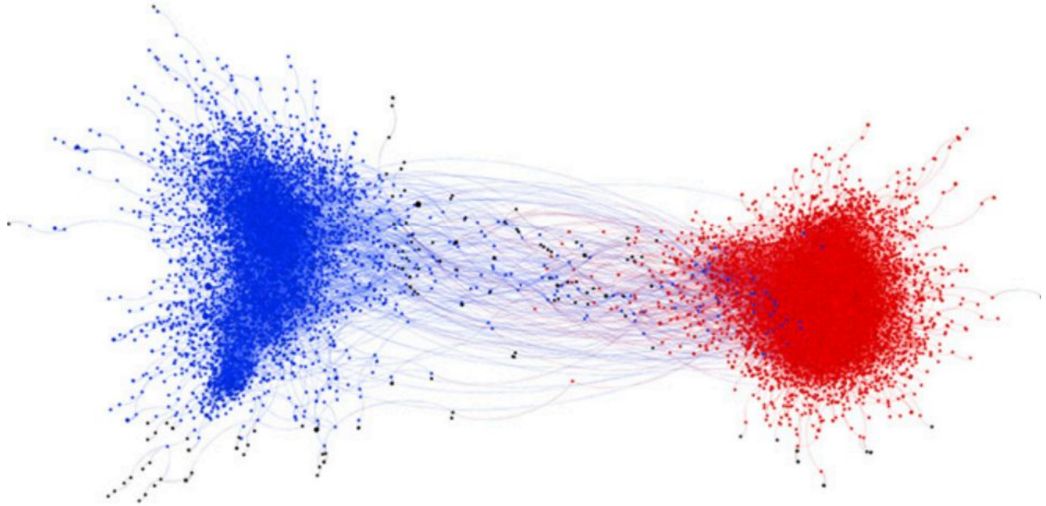


Twitter Emotion Contagion

The effect of moral contagion is greatest within political in-groups compared with political out-groups. Moral-emotional language was associated with a significantly larger retweet rate for the political in-group



Network graph of moral contagion shaded by political ideology. Nodes represent a user who sent a message, and edges (lines) represent a user retweeting another user. The two large communities were shaded based on the mean ideology (blue represents a liberal mean, red represents a conservative mean).



Polarization and Partisan Sharing

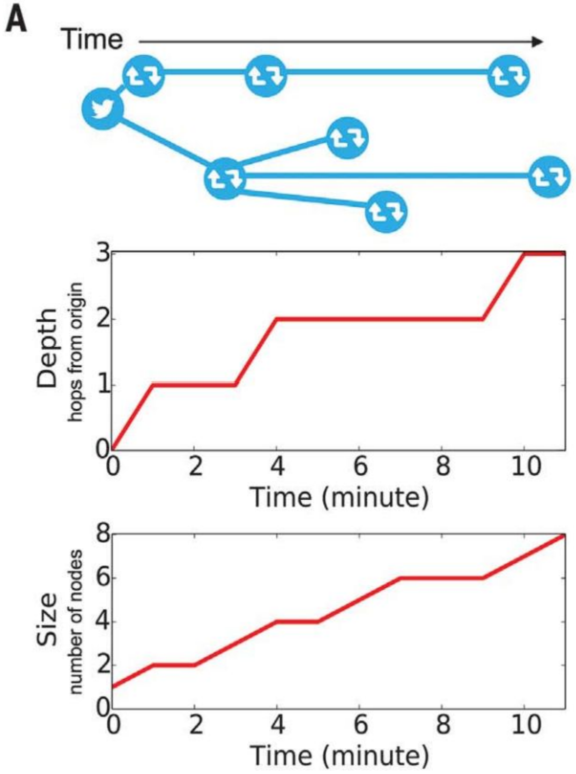
Does emotional and partisan content increase sharing on social media?

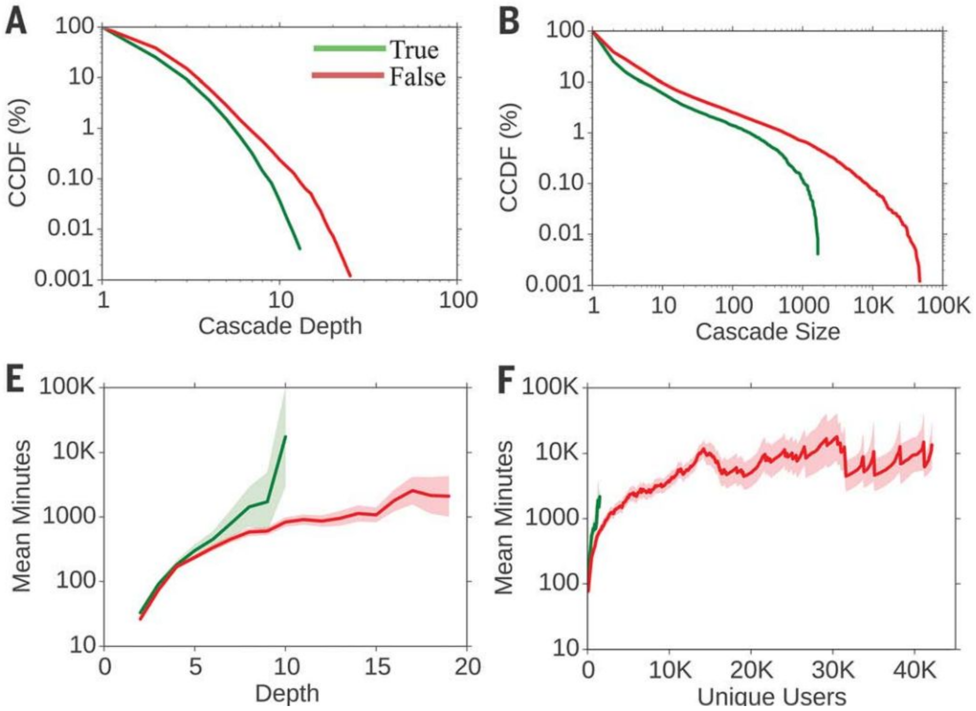
- **Method**:
  - Field experiments on Facebook and Twitter.
  - Users are more likely to reshare content aligned with their partisan identity.
  - Partisan outrage significantly increases resharing, even when users are incentivized to prioritize accuracy.
  - Emotional manipulation fuels political polarization.

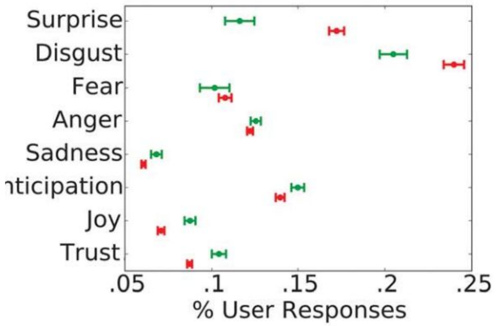
Rumors on Twitter

Vosoughi et al. (2018). The spread of true and false news online. *Science*.

- Rumors (False news) spreads faster and deeper than true news on Twitter.
- Emotional reactions, especially surprise and disgust, drive virality.







How do social media platforms incentivize moral outrage?

[Brady et al. (2021). How social learning amplifies moral outrage expression in online social networks. *Science Advances*]

- **Positive social feedback reinforces outrage (Reinforcement Learning)**
  - Users are more likely to express moral outrage in the future if previous outrage expressions received more likes and shares.
  - The effect is modest but accumulates over time, potentially magnifying at the network level.
- **Social network norms influence outrage (Norm Learning)**
  - Users embedded in **more ideologically extreme networks** express more outrage.
  - This effect persists even when controlling for users' personal ideology.
- **Norms moderate sensitivity to feedback**
  - In extreme networks (where outrage is frequent), users are less sensitive to social feedback. Users internalize expressive norms, making reinforcement learning less necessary.

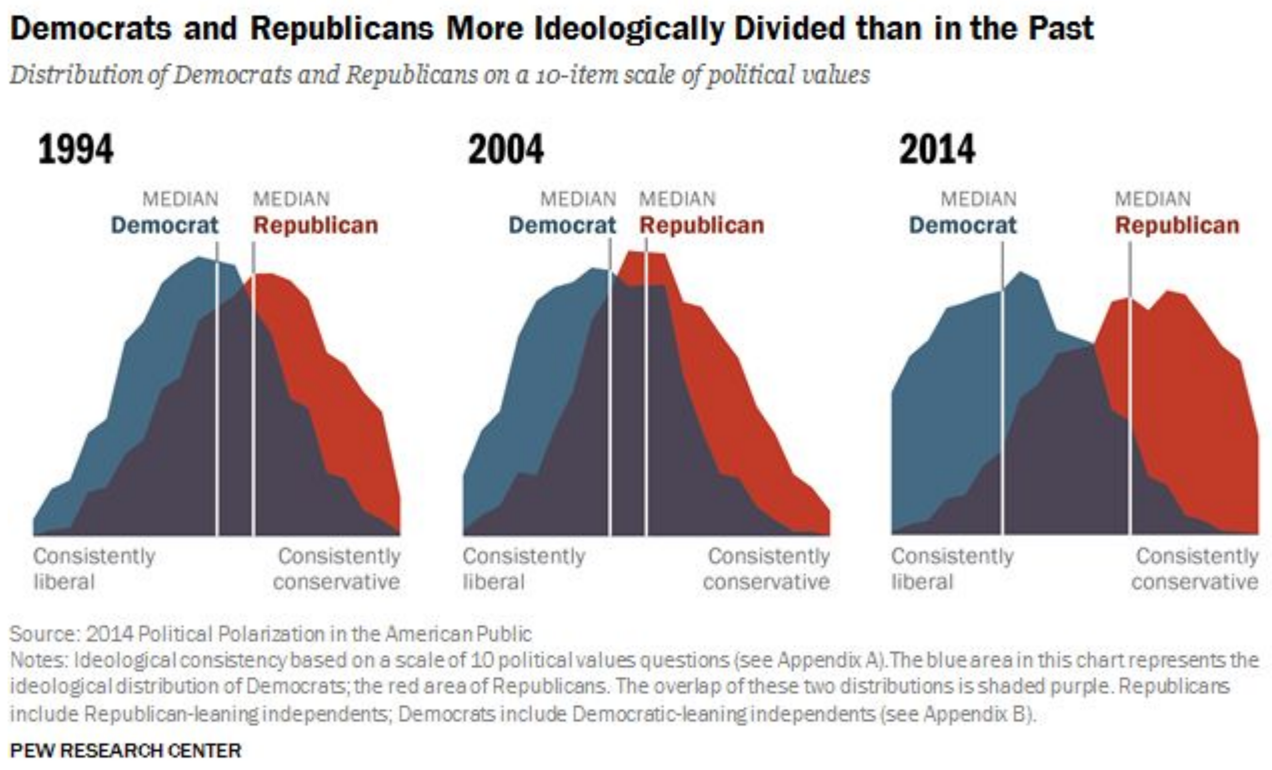
Experimental validation

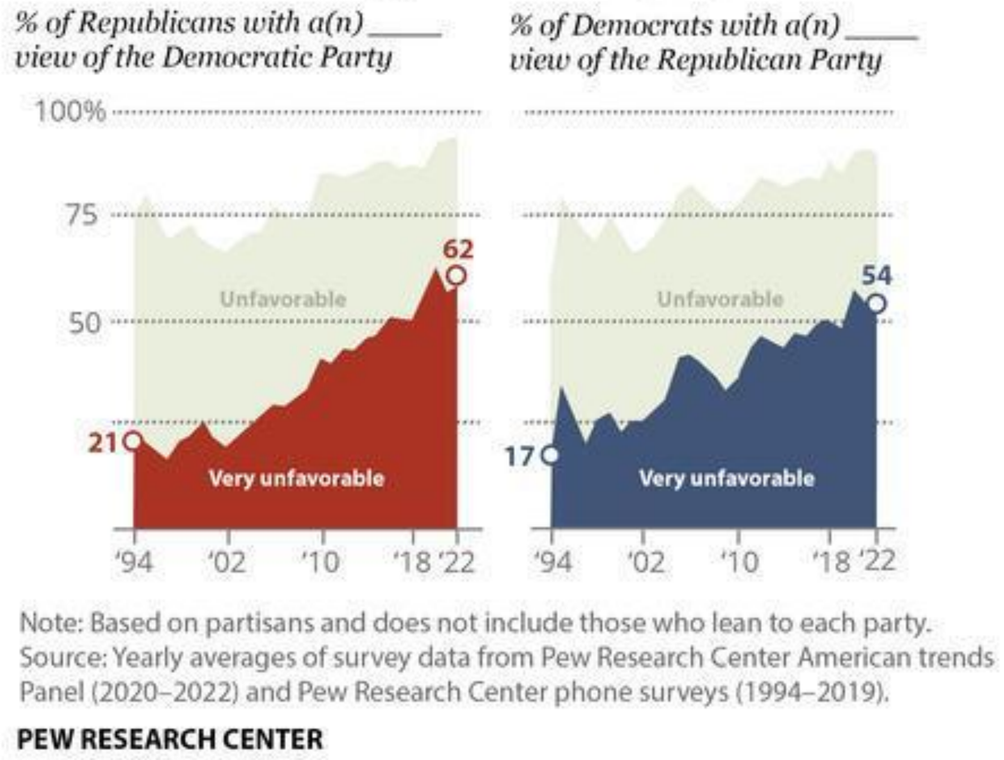
- Two behavioral experiments in a simulated Twitter environment confirmed both reinforcement learning and norm learning.
- In environments where outrage was normative, participants expressed more outrage and relied less on feedback.
- **Platform Design Matters**
  - Social media platforms amplify moral outrage by:
    - Making social feedback (likes/shares) highly salient.
    - Allowing users to form ideologically extreme, homophilic networks
  - Design choices that optimize for engagement can unintentionally escalate moral outrage
- Amplification of outrage may contribute to political polarization and affective polarization

AFFECTIVE POLARIZATION:

how emotional divides create differences of opinion

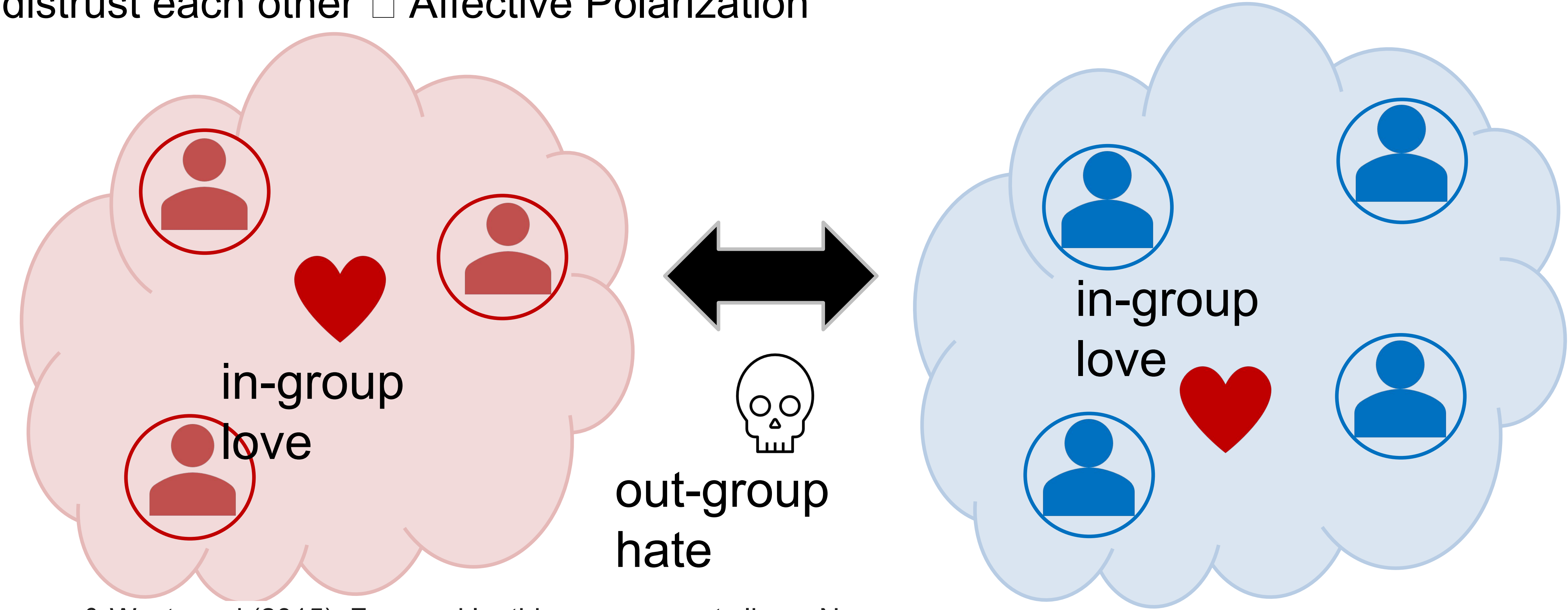
Democrats and Republicans not only disagree but also dislike and distrust each other





Emotional Divides

Different groups not only disagree with each other, but they also dislike and distrust each other □ Affective Polarization



* Iyengar & Westwood (2015). Fear and loathing across party lines: New evidence on group polarization. *American journal of political science*.

Affective polarization on social media

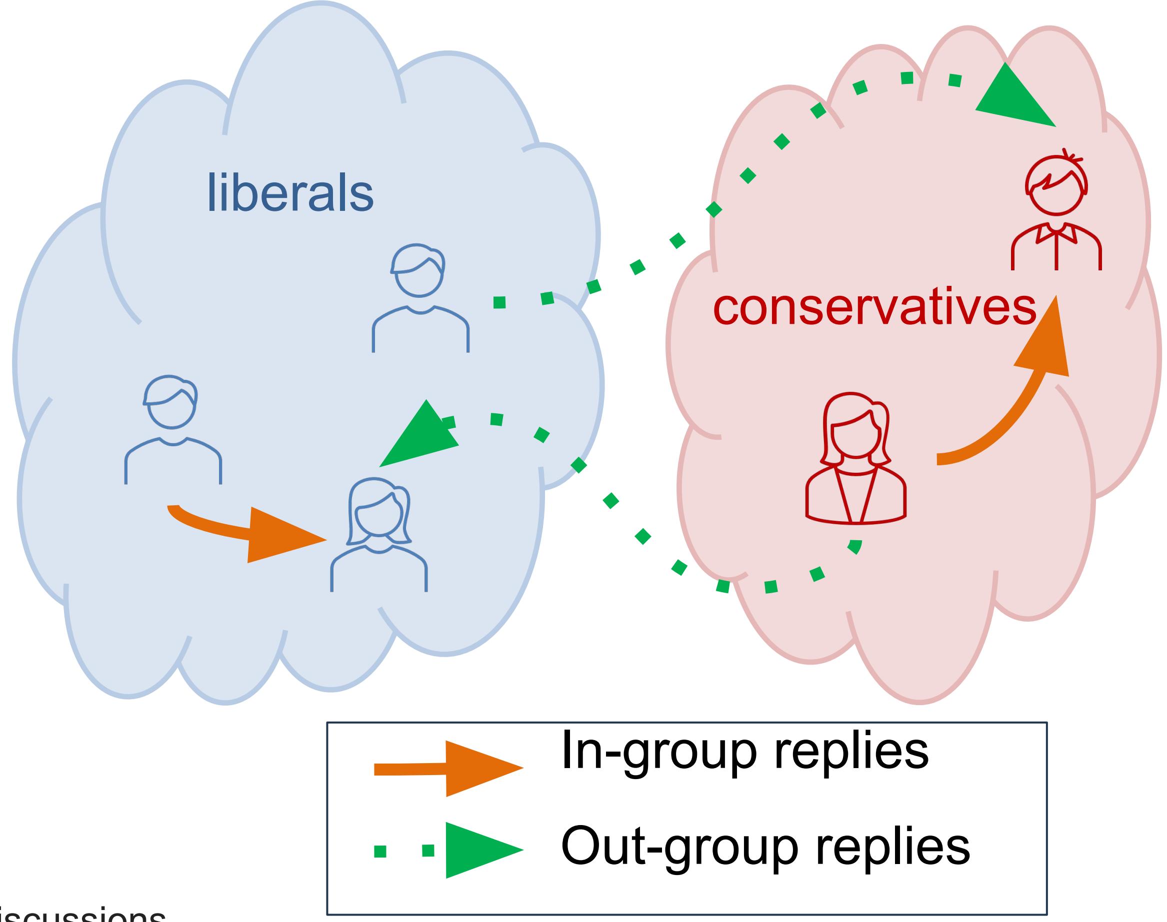
Does affective polarization show up in online interactions?

- In-group love: Positive emotions in replies
- Out-group hate: Negative emotions, toxicity in replies

Social Media Data

- “Roe v Wade” tweets
  - abortion-related keywords
  - 35.7M tweets, 6.4M users
- “COVID-19” tweets
  - pandemic-related keywords
  - 5.2M tweets, 2.2M users
- 2017 French election tweets
  - election-related keywords, tweets in French
- Labeled users as liberal or conservative*

Emotions of reply interactions

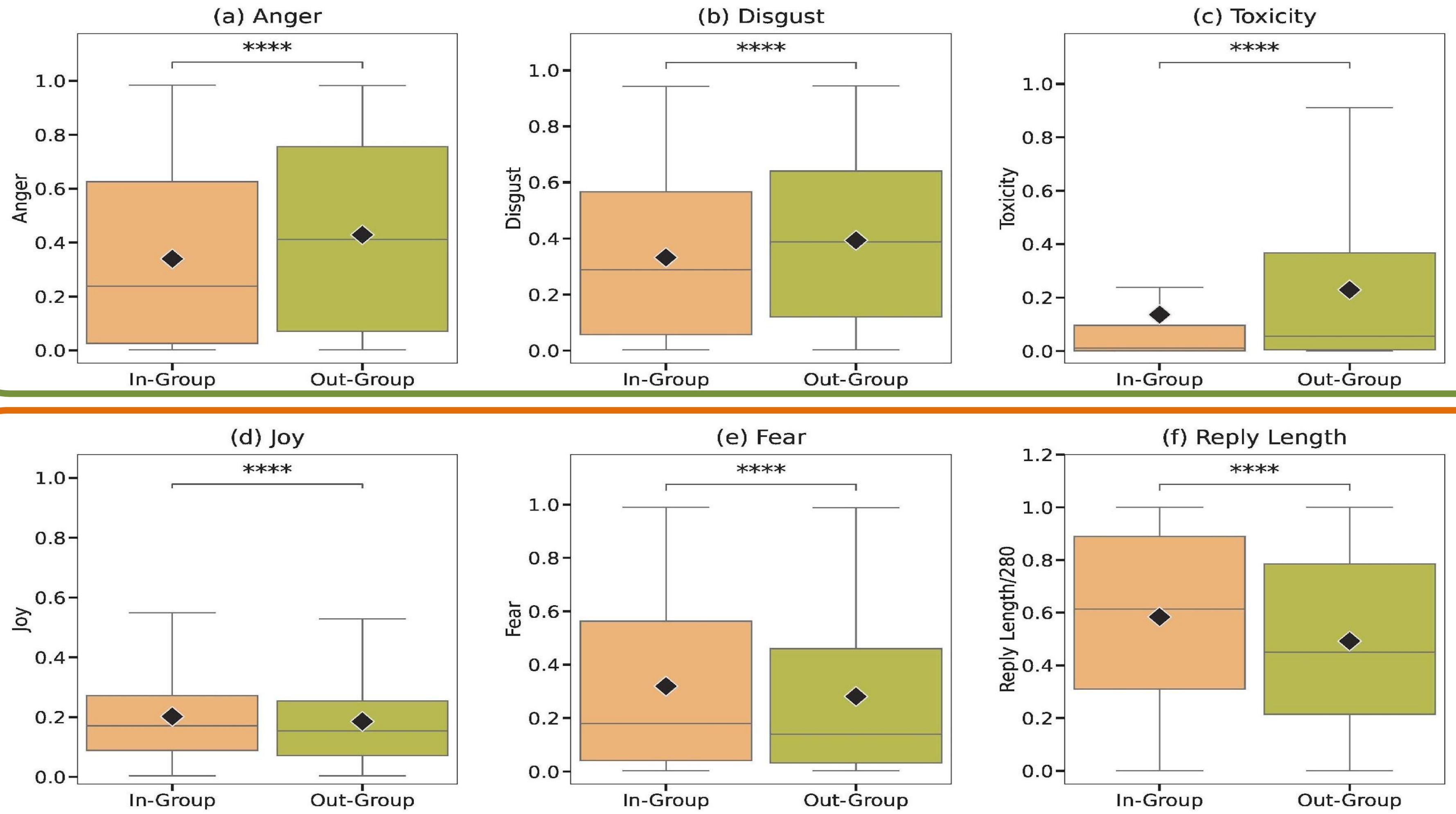


* Rao et al. (2021). Political partisanship and antiscience attitudes in online discussions about COVID-19: Twitter content analysis. *JMIR*, *23*(6), e26692.

Out-group interactions are more negative

Out-group hate

In-group love



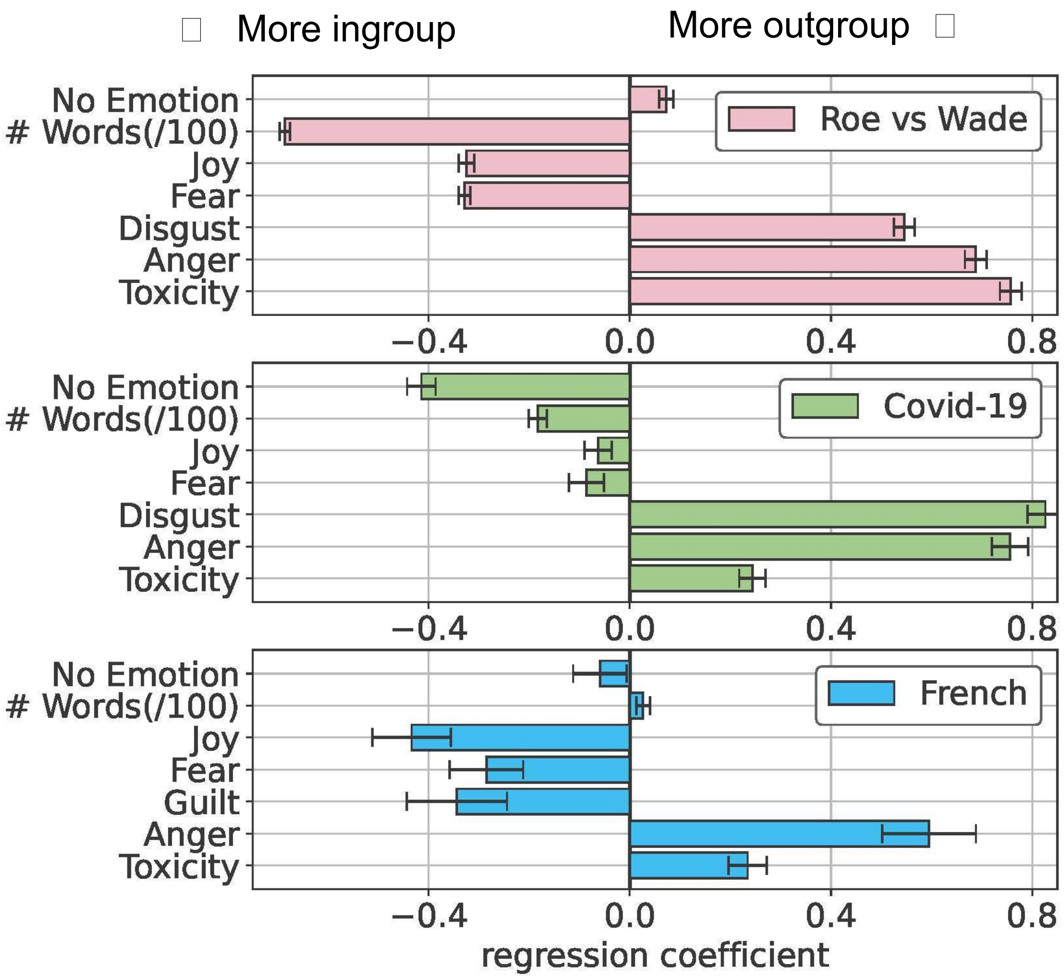
Replies to out-group users are more angry and toxic; less joyful and fearful, and also shorter

Roe vs Wade tweets

Lerman et al (2024). Affective polarization and dynamics of information spread in online networks. *npj Complexity*.

Emotions and groups

Similar trends across data sets, languages, models



People position themselves to be closer to friends and farther from enemies

- Out-group interactions are more emotional
- … more negative
    - Anger, Disgust
    - Toxicity, Obscenity
    - … but Fear, Guilt higher for in-group interactions
        - “honest signal” of group belonging; creates cohesion
- Joy higher for in-group interactions
    - Other positive emotions (love, optimism) no effect

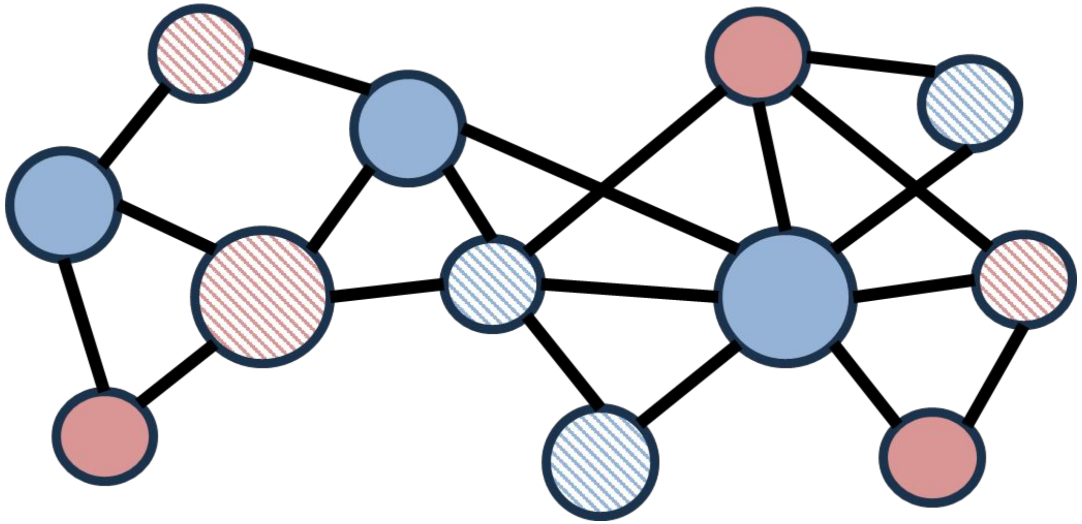
Lerman et al (2024). Affective polarization and dynamics of information spread in online networks. *npj Complexity*.

A model of opinion change with affective polarization

Group (static)

Red (conservative)

Blue (liberal)



Opinion (dynamic)

1 (masks work)

0 (masks don’t work)

- Each node has two binary attributes
  - Static: group affiliation (e.g., red vs blue)
  - Dynamic: opinion (e.g, masks work vs masks do not work)

- Affective polarization
  - ***In-group love*** $\boldsymbol{\alpha}$ drives nodes to conform to opinions of their ingroup (same color nodes)
  - ***Out-group hate*** $\boldsymbol{\beta}$ drives nodes to oppose the opinions of their outgroup (opposite-color nodes)

Nettasinghe, B., Percus, A. G., & Lerman, K. (2024). Dynamics of Affective Polarization: From Consensus to Partisan Divides. *arXiv:2403.16940*.

A model of opinion change with affective polarization

Group (static)

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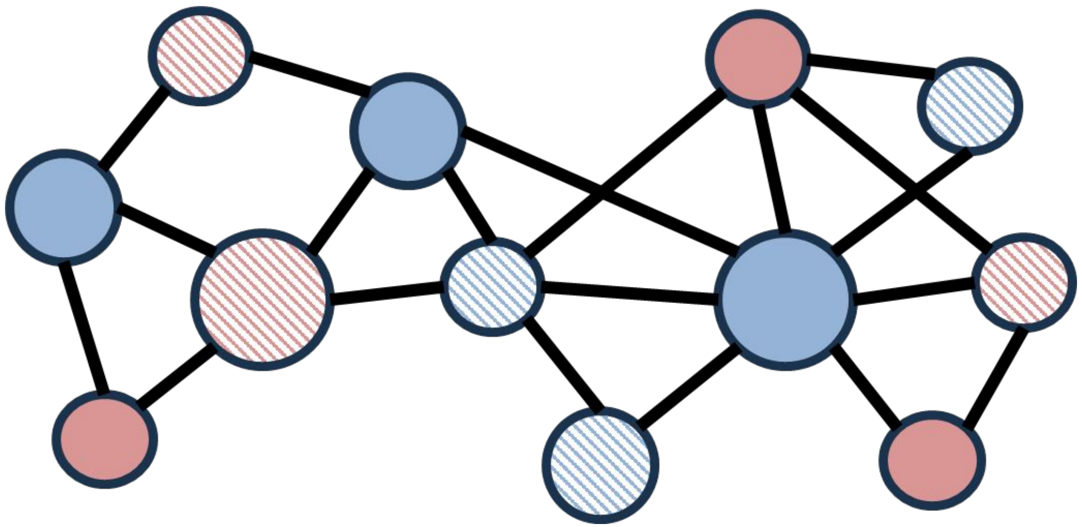


Opinion (dynamic)

1 (masks work)

0 (masks don't work)

Conform to the in-group preference with weight α

Oppose the out-group preference with weight β

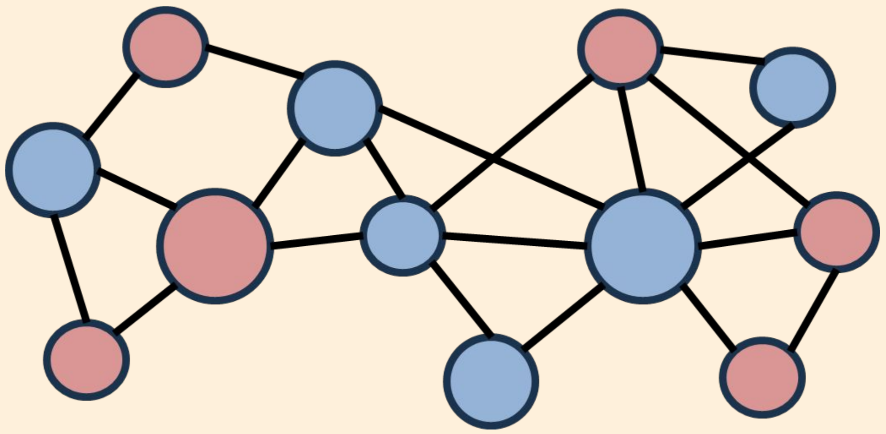
State at time k+1

$$H_{k+1}(X_k) = \begin{cases} 0 & \text{if } \alpha\left(d_k^{in,0}(X_k) - d_k^{in,1}(X_k)\right) - \beta\left(d_k^{out,0}(X_k) - d_k^{out,1}(X_k)\right) > 0 \\ 1 & \text{if } \alpha\left(d_k^{in,0}(X_k) - d_k^{in,1}(X_k)\right) - \beta\left(d_k^{out,0}(X_k) - d_k^{out,1}(X_k)\right) < 0 \\ H_k(X_k) & \text{otherwise,} \end{cases}$$

Steady states of the dynamical system

Global Consensus

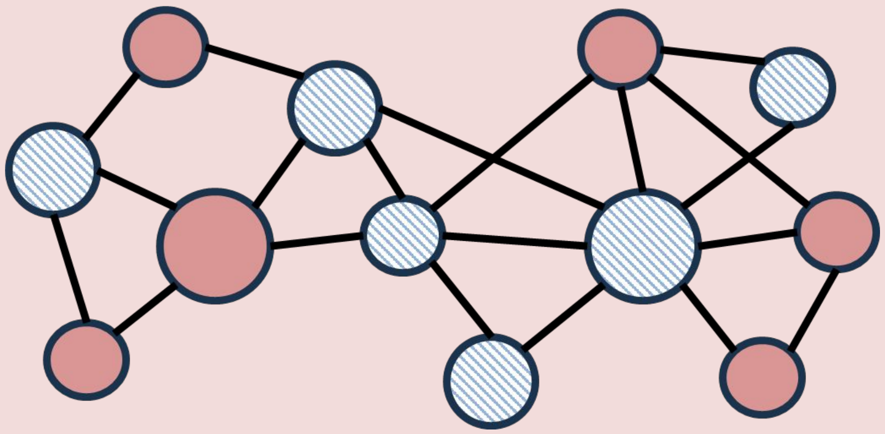
The Red and Blue groups have the same beliefs



○ masks work   ● masks do not work

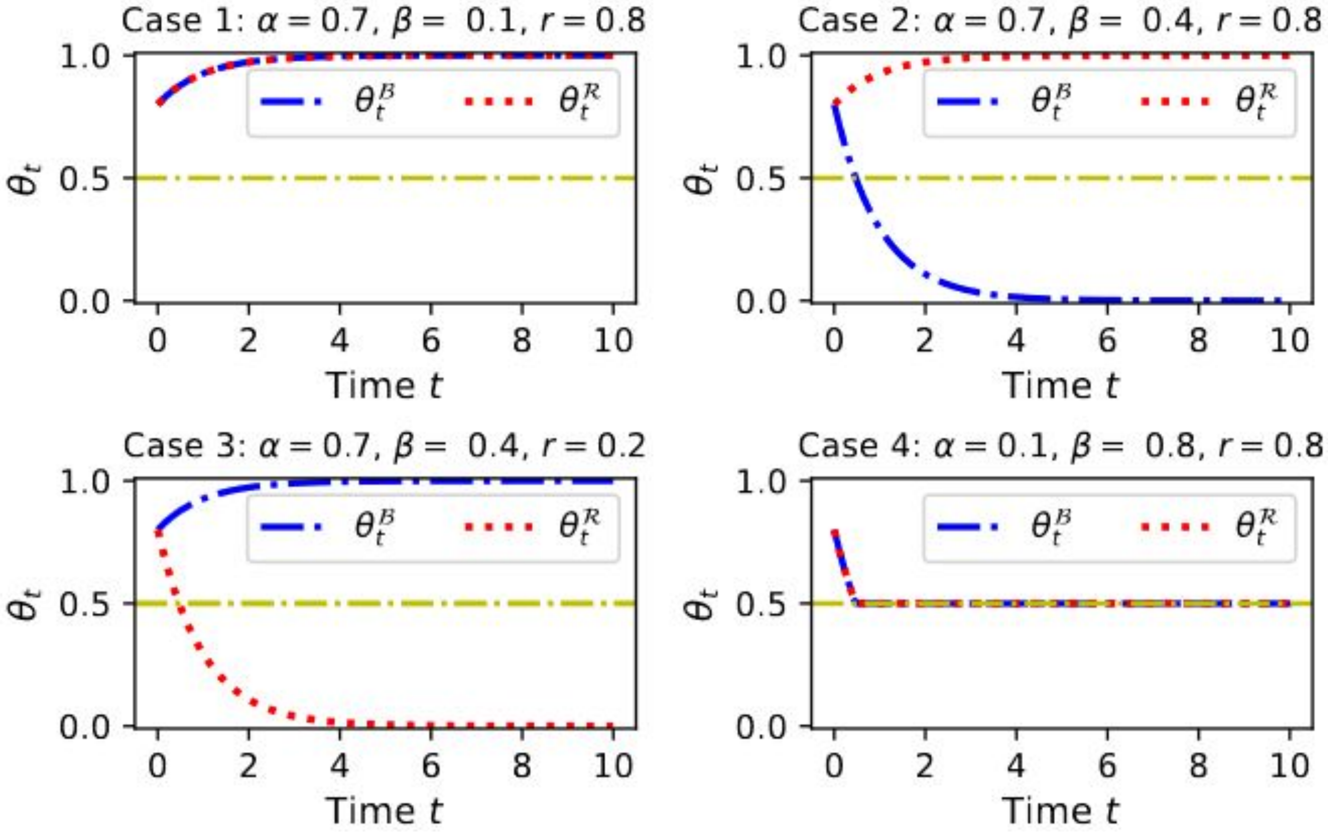
Partisan Polarization

The Red and Blue groups have different beliefs



Steady states of the dynamical system*

θ_t = fraction of the nodes in state 1 (masks work)

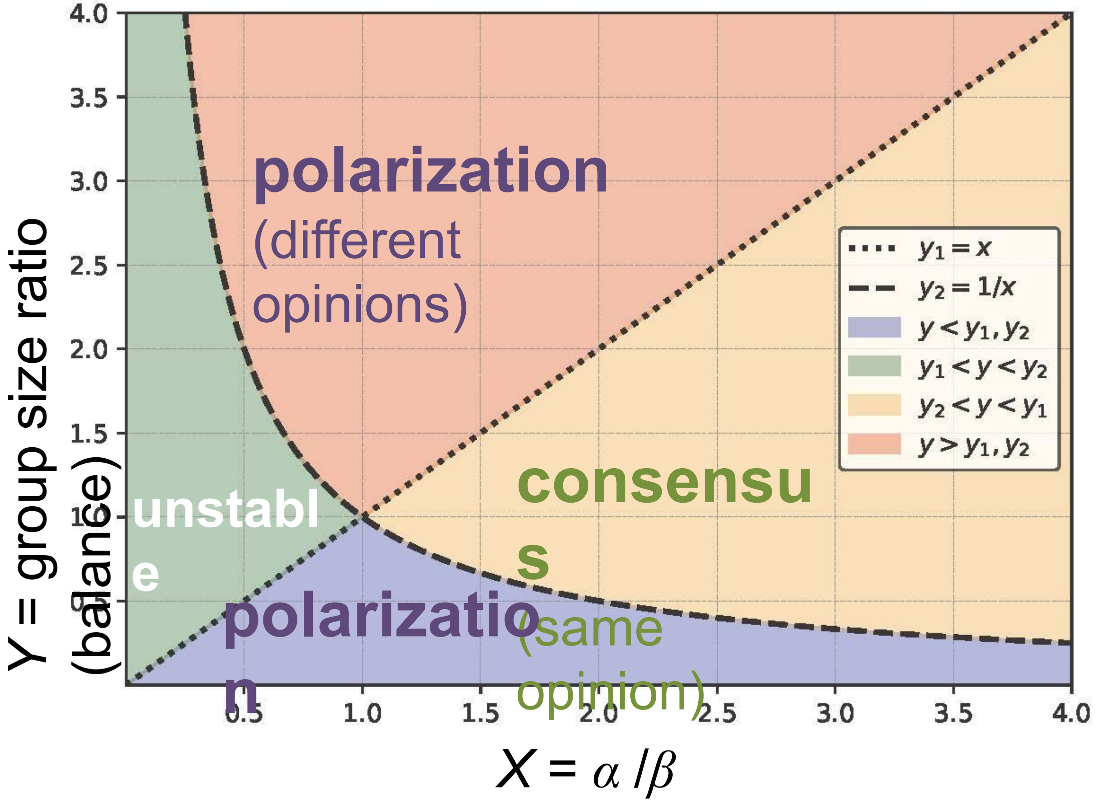


Nettasinghe, B., Percus, A. G., & Lerman, K. (2024). Dynamics of Affective Polarization: From Consensus to Partisan Divides. *arXiv:2403.16940*.

- Case 1: consensus
  - Red is majority (80%)
  - High in-group love (α = 0.7)
  - Low out-group hate (β = 0.1)
- Case 2: polarization
  - Red is majority (80%)
  - High in-group love (α = 0.7)
  - Med out-group hate (β = 0.4)
- Case 3: polarization
  - Red is minority (20%)
  - High in-group love (α = 0.7)
  - Med out-group hate (β = 0.4)
- Case 4: unstable polarization
  - Red is majority (80%)
  - Low in-group love (α = 0.1)

Steady states of the dynamical system*

Phase diagram of the steady state solutions

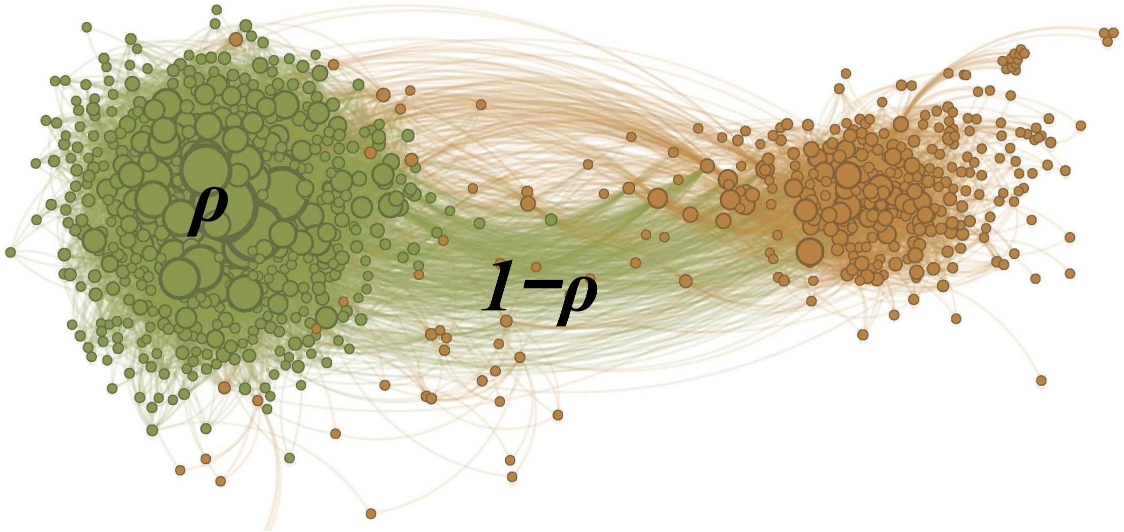


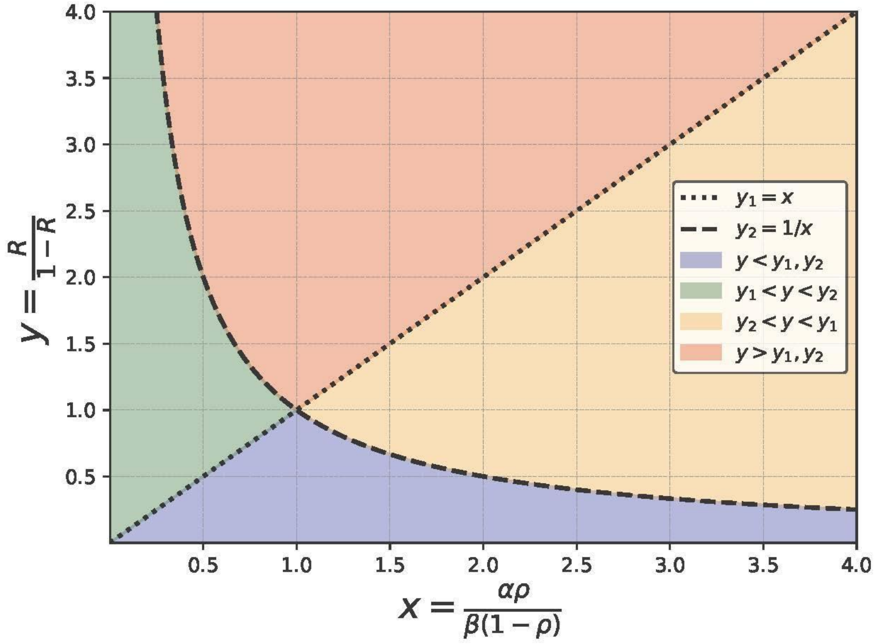
- Regardless of the initial states of red and blue nodes, the system reaches a steady state where the states no longer change
- Final state depends only on
  - X: Ratio in-group love/out-group hate
  - Y: Group size ratio (group balance)
- Sharp boundary emerges between consensus and polarization

* Fully connected system

Affective polarization and echo chambers

- Each node connects to a fraction ρ of same-color nodes and 1-ρ opposite-color nodes (stochastic block model)
  - Homophily ρ
  - Higher homophily ρ □ denser, more isolated communities □ “echo chambers”
  - Similar behavior to fully-connected network
    - Homophily ρ renormalizes in-group love, out-group hat





Breaking echo chambers increases polarization

- Adding connections between groups **increases** polarization
  - Adding links between groups reduces homophily ρ, increases heterophily (1- ρ)
  - High heterophily (1- ρ) amplifies effects of out-group hate
  - Exposing people to out-group views does not decrease polarization

RESEARCH ARTICLE | SOCIAL SCIENCES |

Exposure to opposing views on social media can increase political polarization

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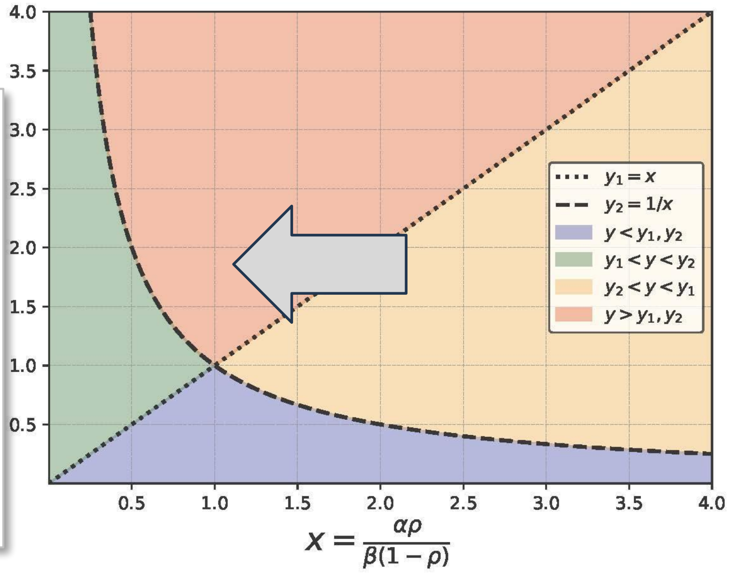
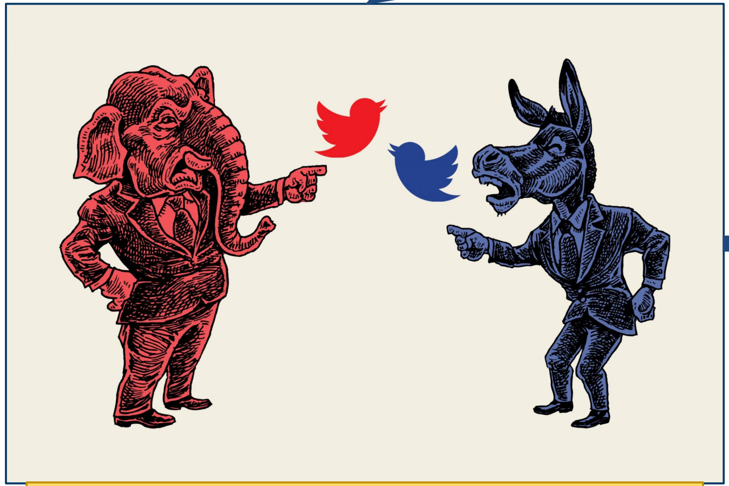


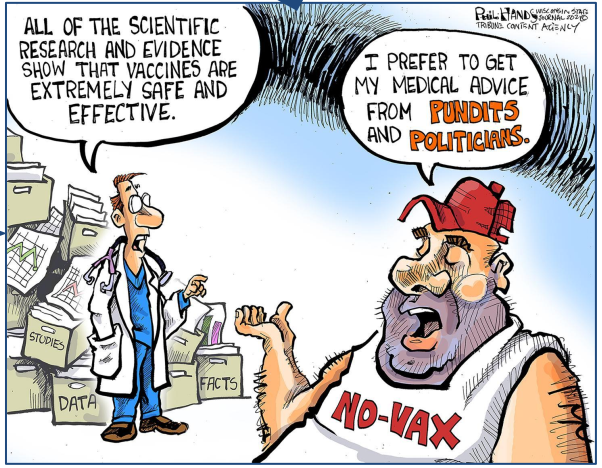
Case study: Rapid polarization during the COVID-19 pandemic

Vicious cycles, esp during crises

Critical events create emotional fault lines



Discourse becomes politicized



Distrust of others

Polarization during the Covid pandemic

- **260M** COVID-19 related tweets [Chen et al. 2020]
- January 21, 2020 to July 31, 2020.
- Less than 1% geolocated.
- Fuzzy geolocation matching [Jiang et al. 2020]
- Roughly **48M** tweets localized to US.

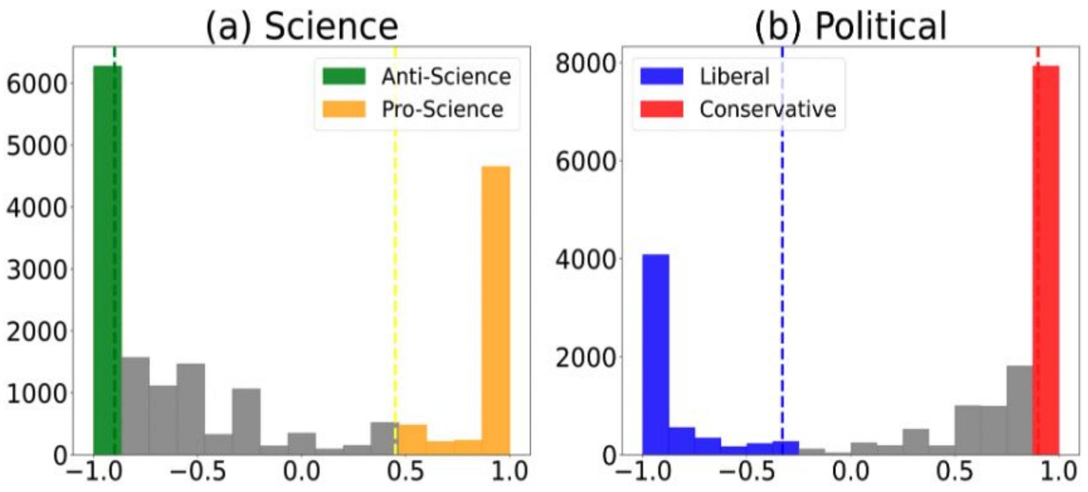
Media Bias / Fact Check

- Ideological bias for news sources.
- **Categories** - Left, Left-Center, Least Biased, Right-Center, Right, Pro-Science, Conspiracy-Pseudoscience, Questionable Sources, Satire.
- **Reporting Quality** - Very High, High, Mostly Factual, Mixed, Low, Very Low
- Over 2K sources with ideological leaning.

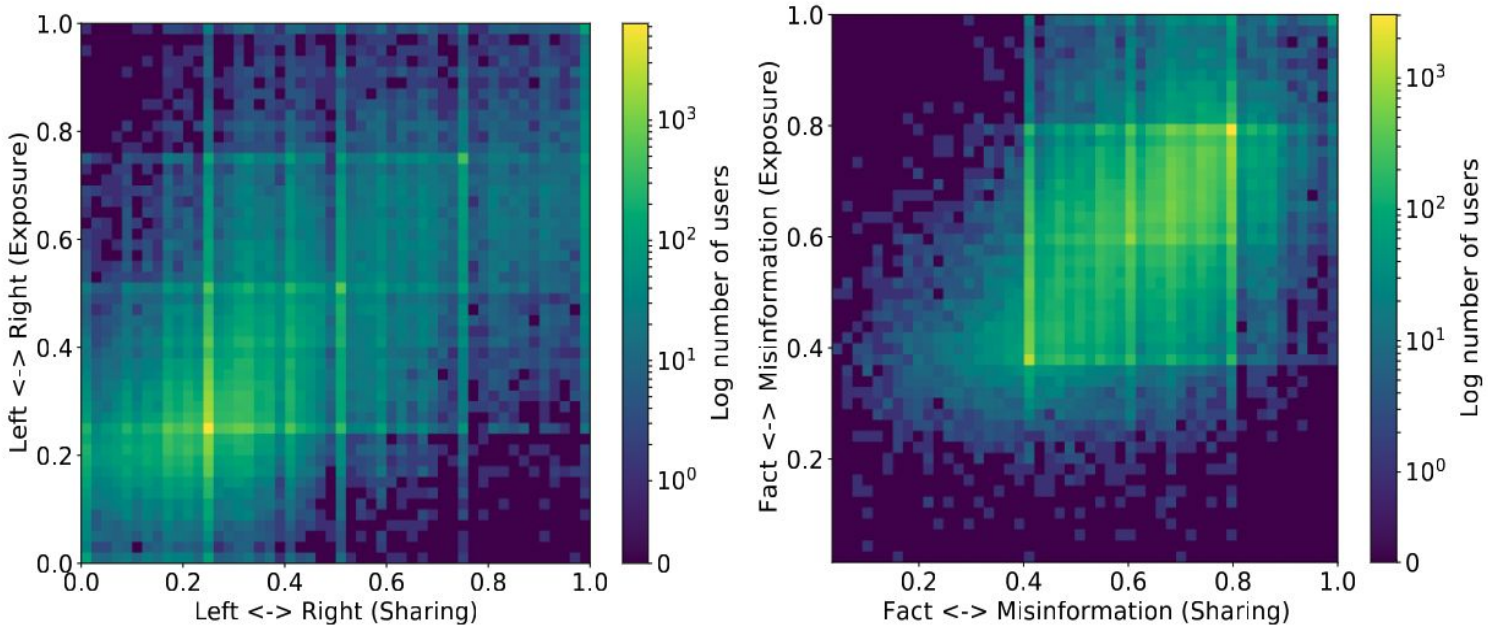
Measuring user political ideology

Content Based Approach
Dimensions: Science (Pro/Anti); Politics (Left/Right);
Assumption: Users share sources that reflect their beliefs.

Dimension	Polarization along dimension	PLDs
Science	Pro-Science (+1)	cdc.gov, who.int, thelancet.com, mayoclinic.org, nature.com, newscientist.com . . . (100+ PLDs)
	Anti-Science (−1)	911truth.org, althealth-works.com, naturalcures.com, shoebat.com, prisonplanet.com . . . (100+ PLDs)
Political	Liberal (−1)	democracynow.org, huffingtonpost.com, newyorker.com, occupy.com, rawstory.com,. . . (250+ PLD s)
	Conservative (+1)	nationalreview.com, news-max.com, oann.com, theepochtimes.com, bluelivesmatter.blue . . . (250+ PLDs)



Echo chambers exist on Twitter



(a) Political Exposure vs Individual Political Scores

(b) Factual Exposure vs Individual Factual Scores

Exposures: What do users see on Twitter?

- Assumption: users see tweets shared by friends
- Collect ***all URLs*** posted by each user's friends
- Compute ideology of friends' tweets (**average exposure**) for each user

- Shows the **joint distribution** of what users ***share*** and what users ***see***.
- **High density diagonals show echo chambers.**
- Much more diffuse than what we see in Cinelli et al., 2021.

Rao et al. Partisan Asymmetries in Exposure to Misinformation. *Scientific Reports* (2022)

Cinelli et al. "The echo chamber effect on social media." *PNAS (2021)*

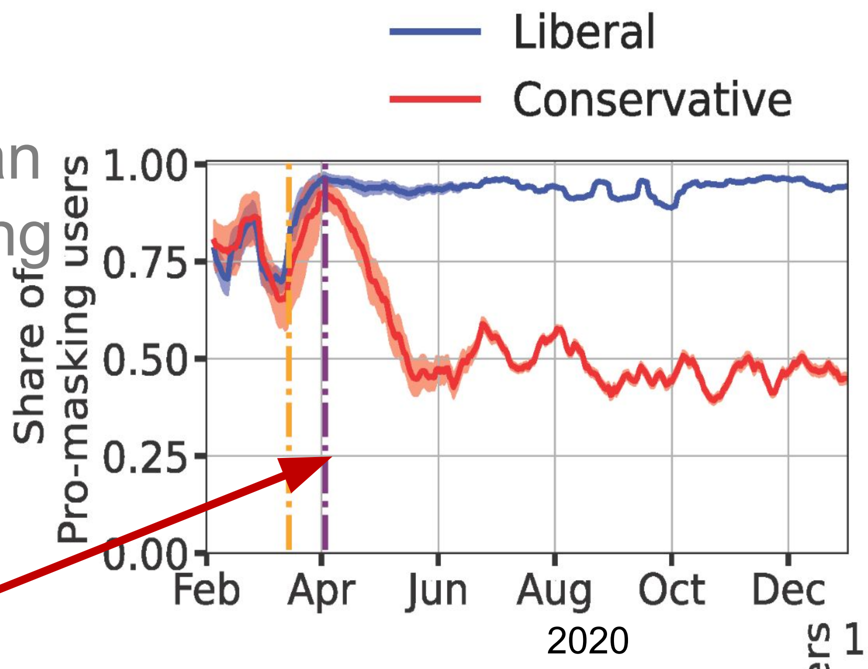
Controversial issues

Education	Masking	Vaccines	Lockdowns	Origins
zoom school, online learning, student, teach…	mask, cloth mask, enforce, face, mandate, …	mRNA, Pfizer, Moderna, vaccination, booster, …	quarantine, stay home, close everything, …	lab leak, wet market, origin, bat, conspiracy, …

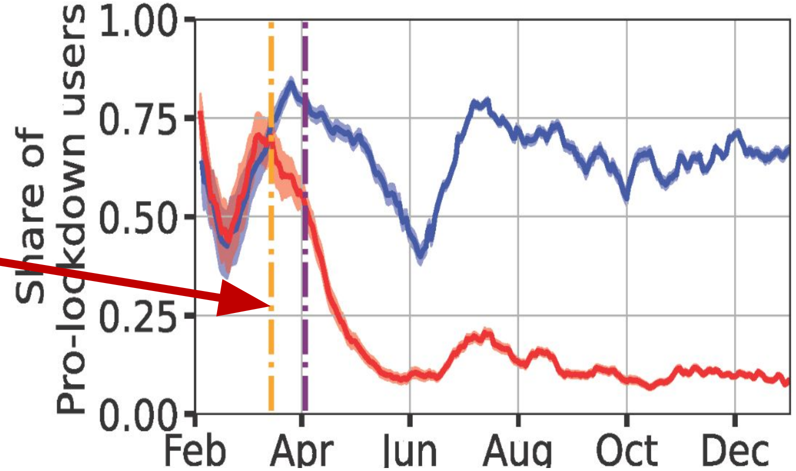
- **Weak supervision framework:** use relevant Wikipedia pages to synthesize training data; train models to identify issues in tweets
- **Focus on US**: 270M tweets from 2.1M users localized to US, through Nov 3, 2021

Rapid polarization during the Covid-19 pandemic

Rapid emergence of a partisan gap in masking attitudes during the COVID-19 pandemic



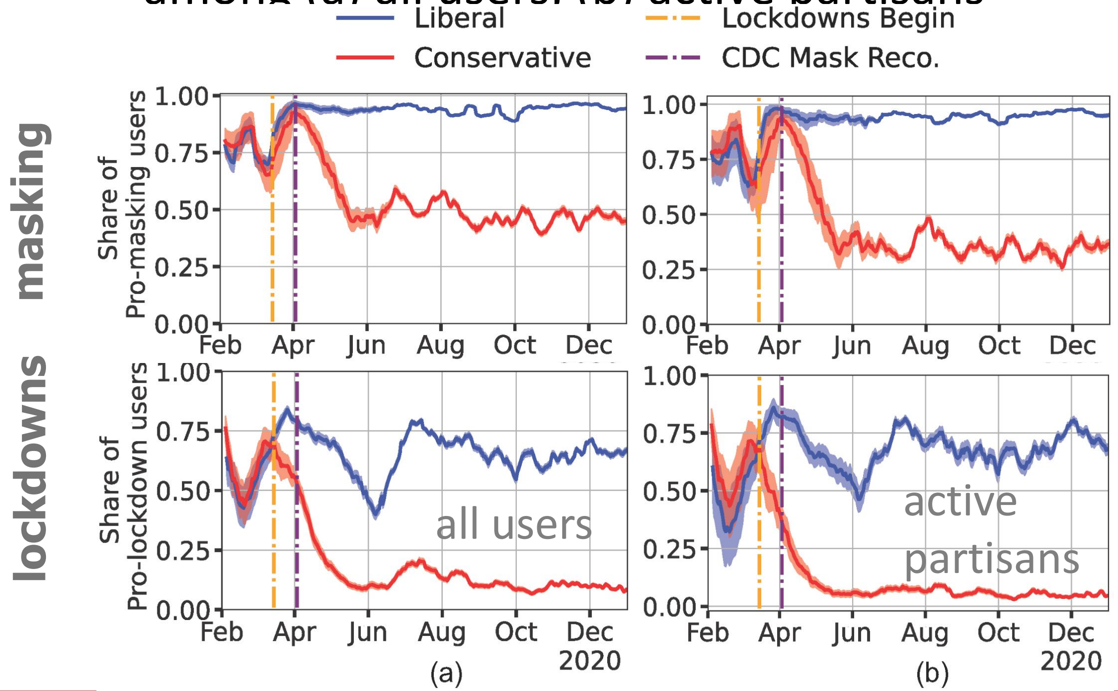
… and lockdowns attitudes



Empirical validation of the model on COVID-19 tweets

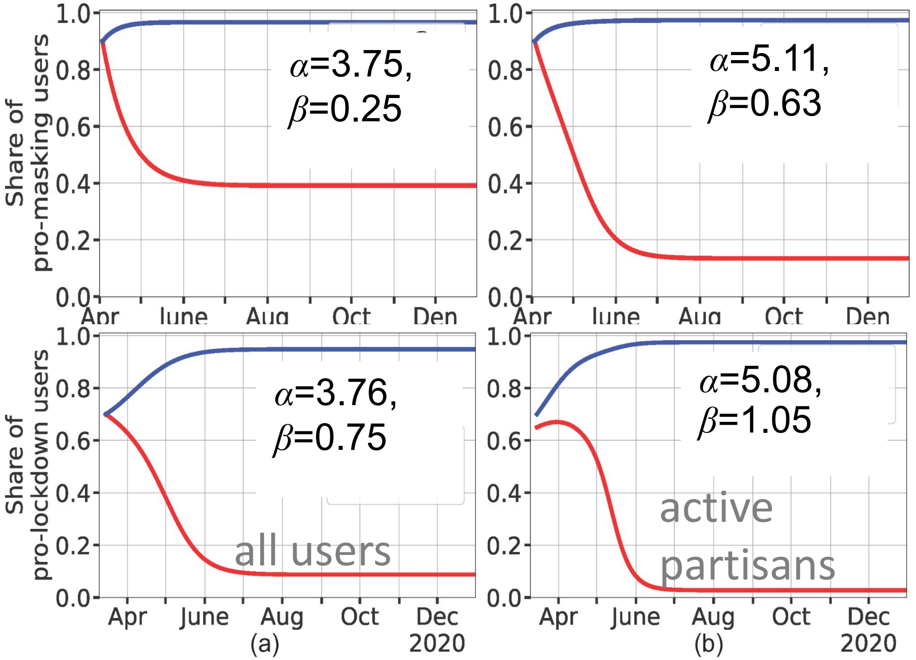
Emotional divide within online networks amplifies polarization on contentious issues

- A partisan gap in the attitudes about masking and lockdowns
- among (a) all users, (b) active partisans



Mathematical model explains rapid polarization

- Estimate in-group love, out-group ($\boldsymbol{\alpha}, \boldsymbol{\beta}$) from data
  - Model replicates emergence of partisan gap from initially similar attitudes with just two estimated parameters



Is social media fueling eating disorders?

Pro-anorexia content is growing rapidly on Twitter

Source: NCRI-Twitter-Report_8.29.pdf

- Pro-anorexia content that glorifies eating disorders is growing rapidly on Twitter
- Users looking for weight loss advice may stumble into pro-anorexia communities
- … and become trapped* among peers endorsing anorexia

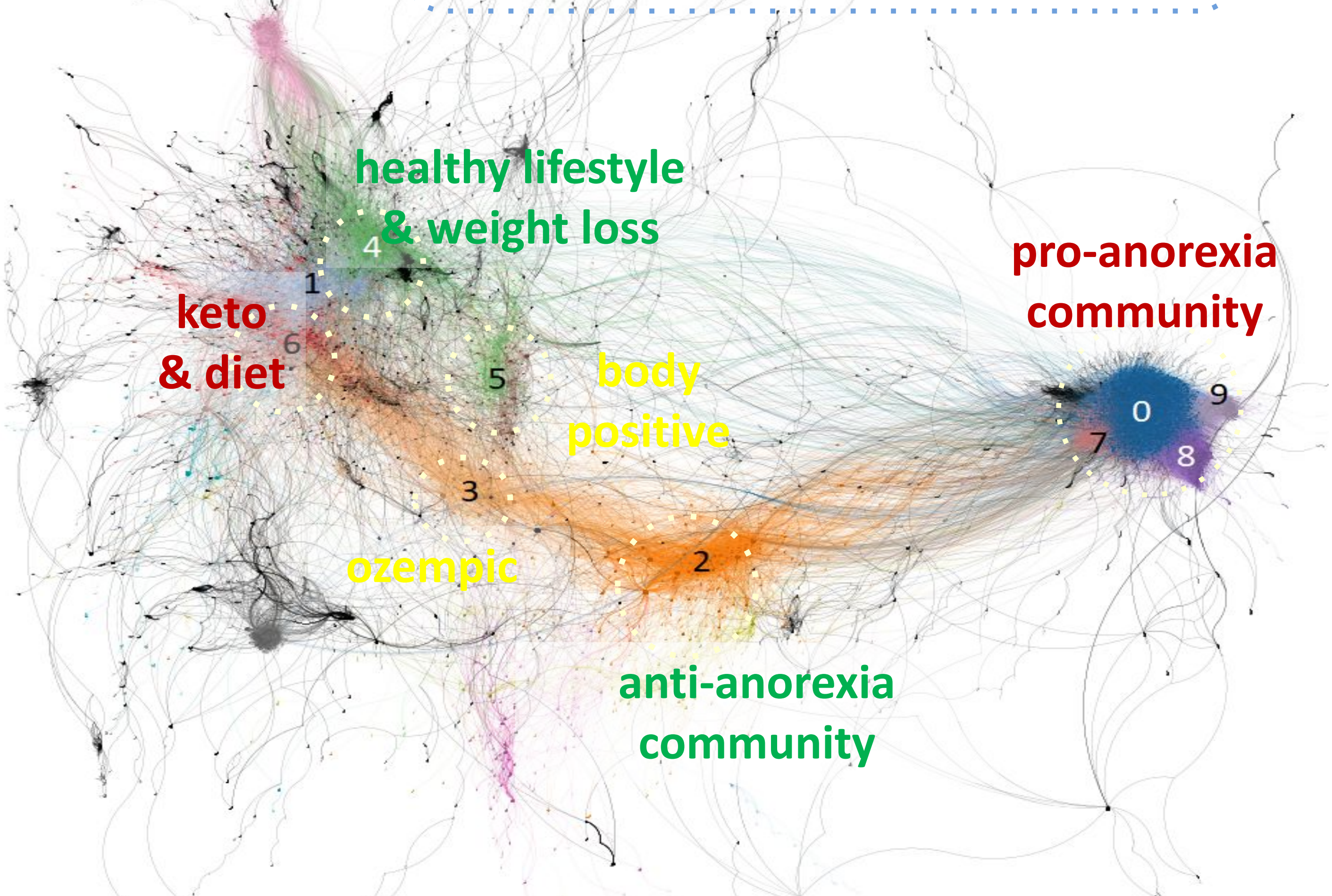
*similar to online radicalization

Eating disorders on Twitter

Retweet interactions: each dot is a user, links between users who retweet each other

3M tweets matching keywords

- Diet & weight loss (keto, ozempic, …)
- Body positivity (dietculture, …)
- Eating disorders (proana, thinspo, …)



RT interactions form communities: Pro-anorexia community forms an echo chamber (few out-group links). **How unhealthy is this community?**

Chu, M. D., He, Z., Dorn, R., & Lerman, K. (2025). Improving and Assessing the Fidelity of Large Language Models Alignment to Online Communities. *NAACL*